
AUTOMATED INTERPRETABILITY METRICS DO NOT DISTINGUISH TRAINED AND RANDOM TRANSFORMERS

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ABSTRACT

Sparse autoencoders (SAEs) are commonly used to learn sparse latent decompositions of transformer activations that are intended to support human-interpretable feature analysis. We evaluate whether standard SAE quality metrics and automated explanation pipelines discriminate between transformers optimized by training and transformers whose parameters are randomly initialized, for example by independent Gaussian sampling. Across Pythia model scales and several randomization procedures, we find that SAEs fitted to randomly initialized transformers frequently obtain auto-interpretability scores and reconstruction-based metrics that are comparable to those obtained from trained models. This observation implies that high aggregate auto-interpretability does not, in isolation, certify recovery of learned or computationally relevant model features. We therefore recommend using common SAE metrics as informative but insufficient proxies for mechanistic interpretability, and we argue for routine randomized baselines together with targeted measurements of feature ‘abstractness.’

1 INTRODUCTION

Sparse autoencoders (SAEs) constitute a standard tool in mechanistic interpretability for decomposing neural-network activations into sparse latent variables that are intended to correspond to interpretable features (Elhage et al., 2022; Sharkey et al., 2022; Cunningham et al., 2023; Bricken et al., 2023). In the usual formulation, an autoencoder with an overcomplete hidden representation is optimized to reconstruct activation vectors subject to a sparsity constraint or penalty (Gao et al., 2024; Templeton et al., 2024; Lieberum et al., 2024); the learned latent coordinates are then treated as candidate concepts or ‘features’ represented by the network (Park et al., 2023; Wattenberg and Viégas, 2024). Progress on SAE methodology therefore depends heavily on quantitative evaluation procedures, including auto-interpretability scores that quantify agreement between generated natural-language explanations and observed activation patterns (Bills et al., 2023; Paulo et al., 2024; Karvonen et al., 2024a).

A robust interpretability evaluation should separate structure acquired through optimization from structure induced by the input distribution, architecture, or measurement procedure. Accordingly, a basic sanity check compares outputs obtained from a trained model with outputs obtained from an appropriate null model, such as a model with randomly initialized weights (Adebayo et al., 2020). We apply this criterion to SAEs and find that several widely used quantitative metrics fail, in some regimes, to clearly separate trained and randomized transformers. Specifically, SAEs trained on activations from transformers with random parameters can produce latent variables whose auto-interpretability scores (Bills et al., 2023; Paulo et al., 2024) are close to those obtained from the corresponding fully trained model.

This finding constrains the interpretation of existing SAE quality metrics. A high auto-interpretability score is not sufficient evidence that an SAE has isolated complex computations acquired during training. The same score may instead be explained by lower-level statistical regularities in the training corpus (Dooms and Wilhelm, 2024) or by architectural inductive biases that remain present before optimization. For example, a randomly initialized network may still implement a minimal transformation that preserves or amplifies sparse structure already present in its inputs (Section 4). Under that interpretation, the SAE may be accurately characterizing an architecture-induced computation rather than a learned mechanism.

Although many SAE latents in trained models plausibly correspond to learned computations, the aggregate metrics considered here do not reliably identify when this stronger condition holds. The implication for mechanistic interpretability is methodological: evaluation protocols must explicitly distinguish metric signal attributable to artifacts from signal attributable to computations induced by training. We therefore advocate validating interpretability methods against carefully specified null models rather than relying exclusively on aggregate SAE quality summaries.

Finally, we analyze two candidate mechanisms that could generate this empirical pattern. At a high level, the hypotheses are: (1) the input distribution already exhibits superposition, and randomly initialized neural networks approximately preserve that superposition; and (2) randomly initialized neural networks amplify, or in some cases introduce, superposed structure in the input representation (e.g., for dense inputs generated i.i.d. from a Gaussian). Section 4 presents toy models that establish the plausibility of these mechanisms, while leaving identification of the dominant mechanism in transformer activations to future work.

2 RELATED WORK

Sparse dictionary learning The phenomenon now described as ‘superposition’ has an antecedent in computational neuroscience, particularly in sparse representations of visual data. Olshausen and Field (1996; 1997) showed that receptive fields of simple cells in mammalian visual cortex can be modeled by sparse coding: a large set of sensory signals is represented through simultaneous activation of a small subset of representational elements. This coding-theoretic viewpoint provides a formal lens for attempts to recover the ‘underlying signals’ that generate neural-network activations (Marshall and Kirchner, 2024).

Sparse dictionary learning (SDL) approximates input vectors as linear combinations of a small number of learned basis vectors. The learned basis is typically overcomplete, meaning that its dimensionality exceeds the dimensionality of the inputs. SDL algorithms include Independent Component Analysis (ICA), which estimates a linear representation whose components are maximally statistically independent (Bell and Sejnowski, 1995; Hyvärinen and Oja, 2000). Sparse autoencoders (SAEs) provide a neural-network implementation of this objective (Lee et al., 2006; Ng, 2011; Makhzani and Frey, 2014). In the standard setting, a single-hidden-layer autoencoder with a hidden dimension many times larger than the activation dimension is trained with an objective that enforces or encourages sparse hidden activations. A **latent** denotes one hidden-layer neuron, equivalently one coordinate of the SAE representation.

Mechanistic interpretability Recent work often formalizes language-model ‘features’ or concepts as low-dimensional subspaces within internal activation spaces (Park et al., 2023; Wattenberg and Viégas, 2024; Engels et al., 2024). If such sparse or ‘superposed’ structure is present, then targeted modifications of the corresponding activation components should support interventions or steering: replacing or perturbing activations to express alternative concepts and thereby alter model behavior (Meng et al., 2022; Zhang and Nanda, 2023; Heimersheim and Nanda, 2024; Makelov, 2024; O’Brien et al., 2024).

SAEs are a common feature-discovery method in which an autoencoder is trained to reconstruct activations from a specified neural-network layer, such as a transformer residual stream layer (Sharkey et al., 2022; Cunningham et al., 2023; Bricken et al., 2023). Proposed SAE architectures frequently differ in the activation or selection rule applied after the linear encoder (Makhzani and Frey, 2014; Gao et al., 2024; Rajamanoharan et al., 2024b; Lieberum et al., 2024). Additional variants modify the training objective (Braun et al., 2024; Farnik et al., 2025) or fit representations across multiple layers simultaneously (Yun et al., 2021; Lawson et al., 2024; Lindsey et al., 2024).

Beyond reconstruction error and preservation of downstream model performance, SAE evaluations test whether learned latents encode specified concepts (Gurnee et al., 2023; Gao et al., 2024), factual knowledge (Huang et al., 2024; Chaudhary and Geiger, 2024), or directions that can be used to ‘unlearn’ concepts (Karvonen et al., 2024b).

Automatic neuron description Because SAEs often contain tens of thousands of latents, exhaustive manual description is generally infeasible. Yun et al. (2021) identify dataset tokens that maximally activate a dictionary element and then manually inspect the resulting activation patterns. More

commonly, recent pipelines collect activation patterns for each latent over a text corpus and prompt a large language model to generate a natural-language explanation (e.g. Bills et al., 2023; Foote et al., 2023). These methods are now widely used in SAE analysis (e.g. Cunningham et al., 2023; Bricken et al., 2023; Gao et al., 2024; Templeton et al., 2024; Lieberum et al., 2024).

Bills et al. (2023) generate natural-language explanations for neuron activation patterns on dataset examples, simulate activations from those explanations, and score an explanation by comparing simulated and observed activations. This protocol is commonly referred to as auto-interpretability, in the sense of self-interpreting models. Paulo et al. (2024) introduce classification-based fidelity metrics for automatic descriptions that are less expensive than activation simulation, together with an open-source pipeline for computing them. Choi et al. (2024) use best-of- k sampling to generate multiple candidate explanations from different subsets of maximally activating examples. They then fine-tune Llama-3.1-8B-Instruct on the top-scoring explanations to obtain lower-cost ‘explainer’ and ‘simulator’ models.

Polysemanticity Lecomte et al. (2024) observe that individual neurons can become polysemantic incidentally. A polysemantic neuron, or basis dimension, represents multiple interpretable concepts within a layer (Elhage et al., 2022; Scherlis et al., 2023); individual neurons in a randomly initialized network may therefore also be polysemantic. Our analysis instead concerns *superposition* (Elhage et al., 2022; Chan, 2024), a property of representations distributed across an entire layer rather than of a single neuron. Under superposition, sparse coding permits a layer to represent more sparse features than it has dense neurons because only a small subset of concepts is active at a given token position.

Training only the embeddings Zhong and Andreas (2024) show that transformers can acquire algorithmic capabilities when training is restricted to the embedding parameters. This result demonstrates that the behavior of an otherwise randomly initialized transformer can be substantially controlled by optimizing a small parameter subset. Our setting differs in two ways: we analyze SAEs, and in the ‘Step-0’ and ‘Re-randomized incl. embeddings’ variants we randomize *all* parameters, including embeddings. The ‘Re-randomized excl. embeddings’ variant retains pre-trained embeddings, but those embeddings are not optimized with fixed randomized weights; they are frozen while the remaining weights are randomized (Section 3).

Random transformers for board games Karvonen et al. (2024c) found that SAEs extracted meaningful structure from chess-game models more effectively when applied to pre-trained transformers than when applied to random-weight transformers. Board-game data differ substantially from language data in the distribution of semantically relevant events. Language plausibly contains sparse concept occurrences, for example rare lexical or semantic concepts such as ‘serendipitous’, and this sparse structure is often aligned with conceptual meaning. By contrast, useful board-game concepts such as a knight fork need not occur sparsely in game records, so the relationship between sparsity and semantic structure may be different.

Random one-layer transformers Bricken et al. (2023) found that auto-interpretability scores discriminated effectively between random and trained one-layer transformers. Our results are compatible with this observation at smaller scales: randomized-model auto-interpretability scores are relatively low for smaller models such as Pythia-70m, while the trained-randomized gap narrows for larger models such as Pythia-6.9b.

3 RESULTS

We fit per-layer SAEs to residual-stream activation vectors from Pythia transformer language models ranging from 70M to 7B parameters (Biderman et al., 2023). For each architecture, we compared SAEs trained on the following variants of the underlying transformer:

- **Trained:** The standard optimized model checkpoint.
- **Re-randomized incl. embeddings:** Every parameter tensor, including embedding parameters, is re-initialized by sampling Gaussian noise whose mean and variance match those of the corresponding trained weight matrix.

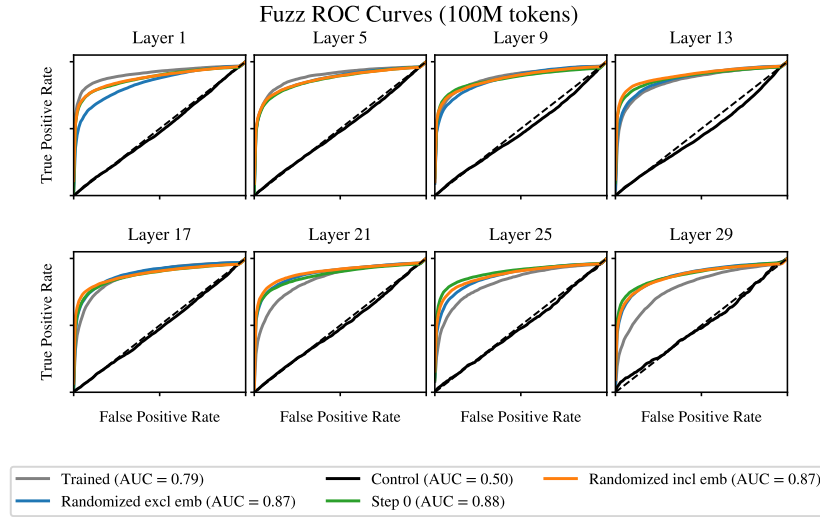


Figure 1: ‘Fuzzing’ ROC curve vs. layer for Pythia-6.9b (100 latents sampled per SAE). The trained model (gray line) and randomized variants (colored) overlap, whereas the control (black) is near chance (dotted). This suggests aggregate AUROC alone is insufficient to attribute latents to learned computation. See Figure 2 for other metrics/model sizes and Appendix E for multiple random seeds.

- **Re-randomized excl. embeddings:** The same re-randomization procedure is applied, except that the embedding and unembedding matrices are retained from the trained model.
- **Step-0:** For Pythia models, the step0 revisions provide the original model weights at initialization, before any learning has occurred (Biderman et al., 2023).
- **Control:** The trained model is evaluated with input token embeddings replaced at inference time by i.i.d. standard Gaussian samples for each token occurrence, so a token ID has no stable embedding vector. We expect auto-interpretability for this variant to be near chance.

In the primary experiments, each SAE was trained on 100M tokens from the RedPajama dataset (Weber et al., 2024) with a 10M-token activation buffer (Appendix C reports a subset of experiments showing similar behavior when SAEs are trained on one billion tokens). For models with fewer than 410M parameters, we trained one SAE at every layer; for Pythia-1b, we trained SAEs at every second layer; and for Pythia-6.9b, we trained SAEs at every fourth layer.

Unless otherwise specified, the SAE architecture was a k -sparse autoencoder (also known as a TopK SAE; Makhzani and Frey 2014; Gao et al. 2024) with expansion factor $R = 64$ and sparsity $k = 32$. We assess robustness to these hyperparameters by training SAEs on Pythia-160m with expansion factors equal to powers of 2 from 16 through 128 and sparsities of 16 and 32 (Figure 18). The training code follows Belrose et al. (2025); the evaluation pipeline follows Caden et al. (2025) and Karvonen et al. (2024a).

Auto-interpretability A generated feature explanation can be evaluated by using it as the specification for a classifier that predicts whether the corresponding concept is present in text inputs. The resulting classifier can be scored with standard binary-classification metrics, including the area under the receiver operating characteristic (ROC) curve (AUROC). Paulo et al. (2024) define ‘fuzzing’ and ‘detection’ classification tasks for evaluating such explanations. Under ‘fuzzing’ scoring, positive and negative token examples for a latent (non-zero and zero activation values, respectively) are marked with special delimiters, and a language model is prompted to decide which examples are correctly delimited given the explanation. Under ‘detection’ scoring, the language model instead predicts which examples contain tokens that activate the feature. Bills et al. (2023) originally proposed ‘simulation’ scoring, which compares predicted and observed activations by correlation, but this procedure is computationally expensive.

Unless otherwise indicated, we use ‘fuzzing’ AUROC as the auto-interpretability measure because it has been shown to correlate with simulation scoring (Paulo et al., 2024). Appendix B reports analogous AUROC curves for the ‘detection’ scoring protocol. For each trained SAE, defined by the underlying model, variant, and layer, we randomly sampled 100 latents for auto-interpretability evaluation. The implementation follows Paulo et al. (2024). We use Meta-Llama-3.1-70B-Instruct-AWQ-INT4 to generate explanations and predictions; this model is larger than the 8B models used by Choi et al. (2024) and is open-source, unlike the model used by Bills et al. 2023.

The resulting auto-interpretability scores for trained and randomized models are substantially closer to one another than to the control scores (Figures 1 and 2). Thus, when applied to SAE latent explanations, ‘fuzzing’ auto-interpretability does not by itself provide a reliable discriminator between trained and randomized transformer backbones.

Evaluation We evaluated standard SAE metrics together with auto-interpretability AUROC for Pythia models ranging from 70M to 6.9B parameters. Consistent with the preceding analysis, the randomly initialized and re-randomized variants (Figure 2; blue, green, orange lines) generally lie closer to the trained variant (Figure 2; gray lines) than to the control (Figure 2; black lines).

The control condition often has substantially lower cosine similarity between the original activation and the SAE reconstruction, lower explained variance, and reconstruction errors that increase rather than decrease across layers. One possible explanation is that, among distributions with fixed mean and variance, the Gaussian has maximum entropy (Jaynes, 2003); in this sense, Gaussian vectors may contain relatively little exploitable structure and may therefore be difficult for SAEs to reconstruct. After Gaussian-distributed activations are propagated through successive layers, the resulting activations may depart from Gaussianity and become more sparse or otherwise more structured, making them easier to reconstruct (Section 4).

The re-randomized variants (blue and orange lines) are also closer to the trained model than to the initialization checkpoint (green line), particularly in the L^1 norm for larger models. We hypothesize that this pattern reflects differences between parameter norms at initialization and after training. By construction, our re-randomization procedure preserves trained-model parameter norms, whereas the step0 checkpoint preserves the original initialization scale. Layerwise parameter scale may affect activation growth through the residual stream (Liu et al., 2020). The AUROC curves further show that, for all non-control variants, AUROC increases with model size. A possible explanation is that larger SAEs allocate latents to more specific activation patterns, whereas smaller SAEs require each latent to account for a broader component of the input, which can make the classification task easier for larger SAEs.

Row five of Figure 2 reports the cross-entropy (CE) loss score, also called loss recovered, as a function of layer. This quantity is the loss increase obtained when original activations are replaced by SAE reconstructions, normalized by the loss increase obtained when those activations are replaced by zeros (‘ablated’). For the trained variant, the resulting SAE performance is comparable to prior reports (e.g., Kissane et al., 2024; Rajamanoharan et al., 2024a; Mudide et al., 2024). The CE loss score is only substantively interpretable for the trained variant: for randomized variants, language-model loss is poor regardless of whether original or reconstructed activations are used.

Latent explanation complexity Although the aggregate metrics can be similar for trained and randomized transformers, we expected the learned SAE latents to differ qualitatively between these settings. In particular, we expected SAEs trained on randomized variants to recover relatively simple features tied to properties of the input text, but not the more abstract features observed in trained transformers (Templeton et al., 2024). Appendix J provides randomly sampled qualitative examples and maximally activating dataset contexts for each Pythia-6.9b variant, with additional dashboard information in Appendix L.

Empirically, many SAE latents appear to activate nontrivially only on one token ID or on a small set of distinct token IDs within a text corpus (e.g., Lin and Bloom, 2024; Dooms and Wilhelm, 2024). This motivates a simple operational proxy for latent complexity: given maximally activating examples, measure whether the latent activation mass is concentrated on one token ID or distributed across many IDs. We quantify this property using the entropy of the observed distribution of latent activation over token IDs; higher entropy corresponds to activation mass spread across more token IDs and therefore to lower token specificity. The distribution is computed as total latent activation

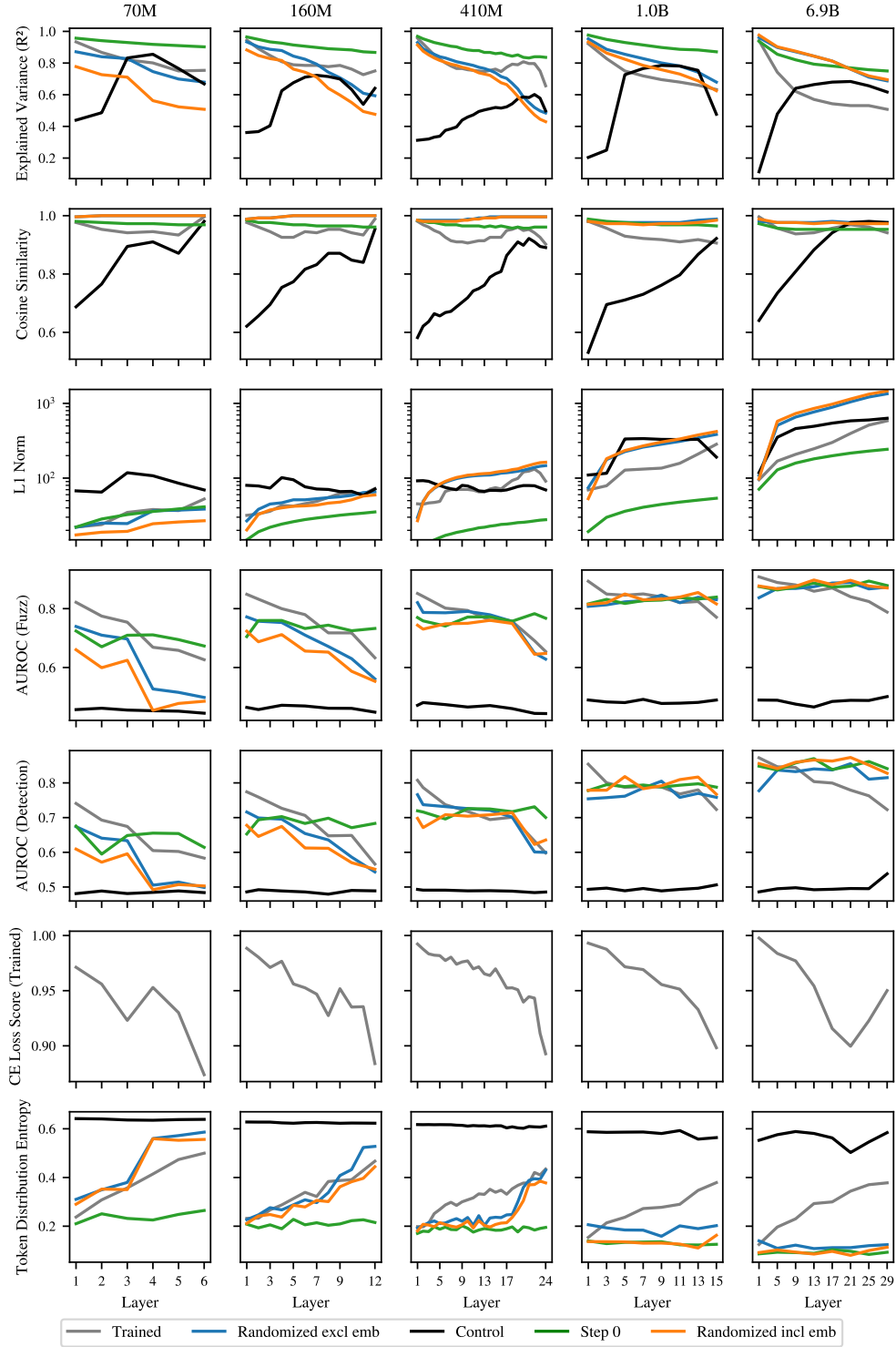


Figure 2: Sparse-autoencoder evaluation metrics across Pythia model scales from 70M to 6.9B parameters. The SAE variants exhibit similar layerwise trends across model scales, and larger models show more consistent behavior across layers. With the exception of the control condition, the variants attain comparable metric values despite substantially different parameter-initialization procedures.

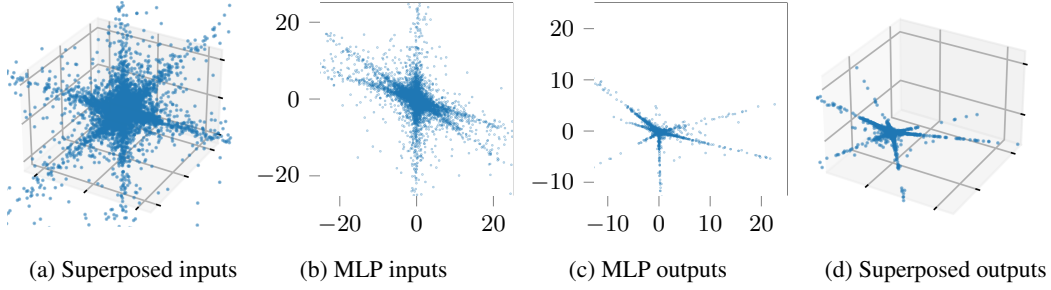


Figure 3: An example of the effect of a randomly initialized neural network on superposed input data. We take 10K samples of $n_s = 3$ sparse input features from a Lomax distribution with shape $\alpha = 1$ and scale $\lambda = 1$ and project these to $n_d = 2$ dense input features by an i.i.d. standard normal matrix. Then, we pass the dense outputs to a two-layer MLP with ReLU activation and hidden size of $4n_d$ and recover $n_s = 3$ sparse outputs by the inverse of the previously generated projection matrix.

per token ID over the maximally activating examples used by the auto-interpretability explanation pipeline. Appendix H reports the per-latent relationship between this entropy measure and ‘fuzzing’ AUROC.

The final row of Figure 2 reports the entropy of the observed token-ID activation distributions. The negative control has consistently high entropy, as expected because each occurrence of a token ID receives a fresh i.i.d. Gaussian embedding and therefore no token ID has a stable embedding vector (Section 3). For the trained variant, entropy increases with layer index: in deeper layers, maximally activating examples for a latent are less likely to concentrate activation on a single token ID. This trend is consistent with the expectation that later layers encode more abstract features that are less directly aligned with token embeddings. By contrast, randomized variants tend to have lower entropy than either the trained or control variants, indicating that their latents often activate specifically on one or a few token IDs.

Together with the preceding metric comparisons, this analysis indicates that standard SAE quality metrics and auto-interpretability scores omit an important property of SAE latents: their degree of ‘abstractness’. Token-distribution entropy is not itself a complete measure of abstraction, but it separates the randomized variants from the trained variant despite their similar aggregate auto-interpretability scores. The randomized variants can learn simple token-specific features that are easy to explain, whereas the trained variant exhibits increasing feature complexity with layer depth.

4 A TOY MODEL OF SUPERPOSITION IN RANDOM NETWORKS

Section 1 identified two possible explanations for the apparent sparsity and interpretability of randomized-transformer activations: the input data may already exhibit superposition that neural networks preserve, or random neural networks may amplify or introduce superposed structure. This section evaluates both mechanisms using toy models. The experiments provide evidence consistent with each mechanism, but determining which mechanism dominates in randomized transformers remains outside the scope of the present study.

4.1 MATRIX MULTIPLICATIONS PRESERVE SUPERPOSITION

We first analyze a simplified generative model showing that multiplication by a weight matrix W preserves superposition. Suppose superposed input data x is generated by sampling n_s i.i.d. sparse features z from a heavy-tailed Lomax distribution, $z \sim \text{Lomax}(\alpha, \lambda)$. A projection matrix D maps the higher-dimensional sparse vector z into a lower-dimensional dense representation x , after which small-variance Gaussian noise with covariance Σ is added: $x \sim \mathcal{N}(x; Dz, \Sigma)$. If x is multiplied by a matrix W , the transformed vector $x' = Wx$ remains superposed under the same model class, with the projection and noise covariance transformed to WD and $W\Sigma W^T$, respectively: $x' \sim \mathcal{N}(z; WDz, W\Sigma W^T)$.

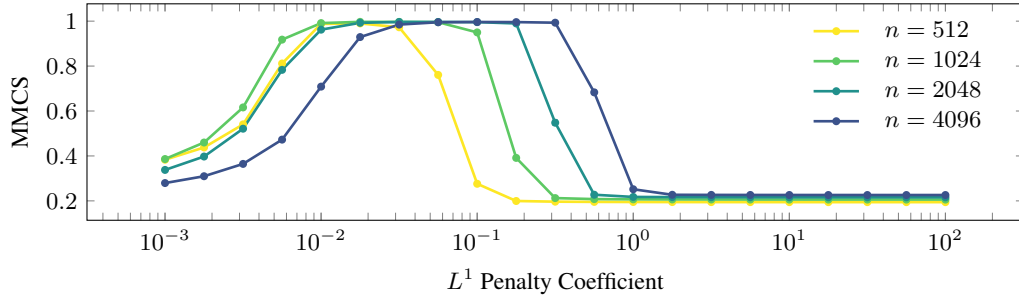


Figure 4: Mean max cosine similarity (MMCS) between standard-SAE decoder weight vectors and the data-generating features as a function of the L^1 penalty coefficient in the training loss, following Sharkey et al. (2022). For an intermediate range of penalty values, SAEs nearly recover the data-generating features when the latent dictionary has sufficient capacity.

A related intuition can be examined for nonlinear networks by passing the dense activations through a simple feed-forward MLP. Figure 3 illustrates a case in which $n_s = 3$ sparse features are projected to $n_d = 2$ dense features, and the MLP outputs still exhibit visually superposed structure despite the nonlinearity. The visualization also suggests possible amplification: relative to the inputs (Figures 3a and 3b), the outputs (Figures 3d and 3c) contain fewer points between the apparent ‘arms’.

4.2 DO RANDOM NNs PRESERVE OR AMPLIFY SUPERPOSITION?

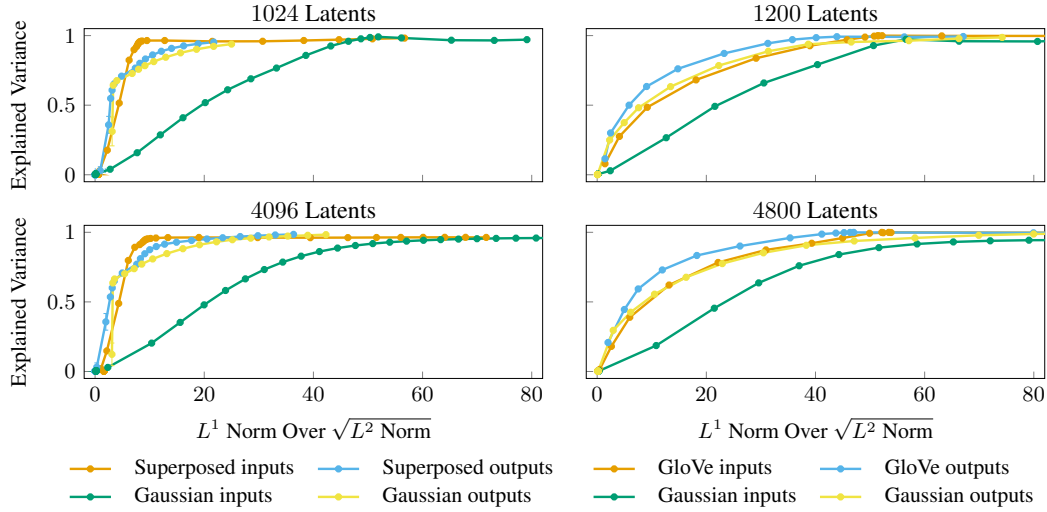
We tested this hypothesis by generating toy data using the procedure of Sharkey et al. (2022): ground-truth features are sampled on a hypersphere, and correlated feature coefficients are generated so that only a small subset is active in each sample (Appendix I.1). The resulting inputs were passed through a randomly initialized two-layer MLP, and separate SAEs were trained on the inputs and on the MLP outputs. The control dataset consisted of Gaussian inputs with mean and standard deviation matched to the superposed toy data. All toy experiments used standard SAEs with an L^1 sparsity penalty (Appendix I.2).

Following Sharkey et al. (2022), we first verified that SAEs can recover the ground-truth features used to generate the toy inputs (Figure 4). The recovery metric is mean max cosine similarity (MMCS): for each data-generating feature, we compute the maximum cosine similarity with any learned SAE decoder vector and then average over data-generating features. Because MMCS requires access to ground-truth features, it is only applicable to the MLP inputs and cannot directly evaluate the MLP outputs. We therefore use the ability of SAEs to obtain low reconstruction error at high sparsity as a proxy for the degree of superposition in a dataset. Operationally, we vary the L^1 penalty coefficient and plot Pareto frontiers of explained variance against sparsity measures (Figure 5a).

As expected, for a fixed explained-variance level, SAEs achieve substantially higher sparsity on superposed inputs than on the Gaussian control (Figure 5a; orange and blue-green). After the inputs are passed through the randomly initialized MLP, however, the gap between outputs derived from superposed inputs and outputs derived from Gaussian inputs is much smaller. Both output distributions support relatively sparse SAE reconstructions, suggesting that randomly initialized MLP outputs can exhibit substantial apparent sparsity with limited dependence on the input distribution. Appendix I reports additional sparsity measures and hyperparameter settings.

4.3 DO TOKEN EMBEDDINGS EXHIBIT SUPERPOSITION?

If randomly initialized neural networks preserve or amplify superposition, then the main results (Section 3) may depend on the extent to which transformer inputs already exhibit superposed structure. We evaluate this possibility by applying the procedure from Section 4.2 to language-derived representations. Specifically, we train SAEs on pre-trained GloVe word vectors, Pythia embedding matrices, the outputs obtained by passing these representations through a randomly initialized two-layer MLP, and matched Gaussian controls. The experimental setup is the same as in Section 4.2, except that dataset size is determined by the number of word embeddings or token embeddings and only one random seed is used.



(a) Toy datasets following Sharkey et al. (2022). (b) 300-dim. GloVe vectors (Pennington et al., 2014).

Figure 5: Pareto frontiers of explained variance against the L^1 norm divided by the square root of the L^2 norm (sparsity) for datasets that may exhibit superposition, Gaussian controls matched in mean and variance, and the corresponding outputs produced by a randomly initialized two-layer MLP (Section 4.2). Each point corresponds to one L^1 penalty coefficient.

The Pareto-frontier gap between GloVe word vectors and their matched Gaussian controls (Figure 5b) is smaller than the corresponding gap for the toy superposed datasets in Section 4.2 (Figure 5a). Nevertheless, passing both inputs through a randomly initialized two-layer MLP improves the Pareto frontiers, reinforcing the possibility that random NNs ‘sparsify’ inputs by increasing apparent superposition.

5 LIMITATIONS

This work shows that auto-interpretability measures can assign apparently meaningful explanations to SAEs trained on randomly initialized models, even though such models are unlikely to contain learned computations of the kind typically targeted by mechanistic interpretability. Because an exhaustive sweep over datasets and architectures is infeasible, we restrict the empirical study to the Pythia model family, which is widely used in mechanistic-interpretability research (e.g., Paulo and Belrose, 2025; Ghilardi et al., 2025; Mueller, 2024), and to RedPajama-V2, which is representative of pre-training data used for language models and SAEs.

We use the default explanation model from the EleutherAI auto-interpretability framework (Caden et al., 2025); evaluating alternative explanation models may change explanation quality or aggregate score calibration and is an important direction for follow-up work. We also do not claim that SAEs fail to capture information specific to trained Transformers. The narrower conclusion is that aggregate auto-interpretability scores do not, by themselves, establish that the recovered latents correspond to computationally interesting learned features.

6 CONCLUSION

We trained sparse autoencoders on activations from both trained and randomly initialized transformers and evaluated the resulting latents using common quantitative SAE metrics. The central empirical result is that, in several regimes, these metrics—especially aggregate auto-interpretability scores—take similar values for trained and randomized backbones. At the same time, qualitative examples and token-distribution entropy indicate that latents from trained transformers can become more complex and abstract, particularly in later layers, even when the aggregate metrics do not reflect that distinction.

486 These findings should not be interpreted as evidence that SAEs trained on real models fail to
487 recover meaningful computational features. Instead, they identify a limitation of current evaluation
488 protocols. High aggregate auto-interpretability can be produced by structure already present in
489 the data distribution or induced by the architecture, and therefore is not sufficient evidence for
490 recovery of complex learned computation. The token-distribution entropy analysis is a preliminary
491 proof of concept: it detects differences in feature ‘abstractness’ that aggregate auto-interpretability
492 scores obscure. Future evaluation methods should more directly quantify the computational role and
493 abstraction level of SAE latents. More broadly, interpretability methods should be benchmarked
494 against strong, explicitly specified null models, such as the randomized transformers used here,
495 because otherwise it is difficult to attribute a discovered feature to training rather than to data or
496 architecture alone.

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A BROADER IMPACT

This work evaluates a method currently used for mechanistic interpretability of LLMs and identifies a failure mode of common sparse-autoencoder evaluation practices. By showing that SAEs can obtain similar aggregate auto-interpretability scores on random and trained transformers, the results constrain what these metrics can be interpreted as measuring.

A more precise understanding of SAE quality metrics may support the design of improved SAE-like methods and, in turn, more reliable interpretability tools for mitigating harms from language models. Because the contribution is an empirical evaluation of an existing method, and because it primarily cautions against over-interpreting metric values, we expect the direct risk of negative social impact to be minimal.

B AUTO-INTERPRETABILITY ROC CURVES

Figures 6, 8, and 12 report ‘fuzzing’ AUROC for trained and randomized SAEs on the 70M, 160M, and 1B Pythia models. Figures 7, 9, and 13 report the analogous ‘detection’ AUROC comparisons for the same model scales.

B.1 PYTHIA 70M

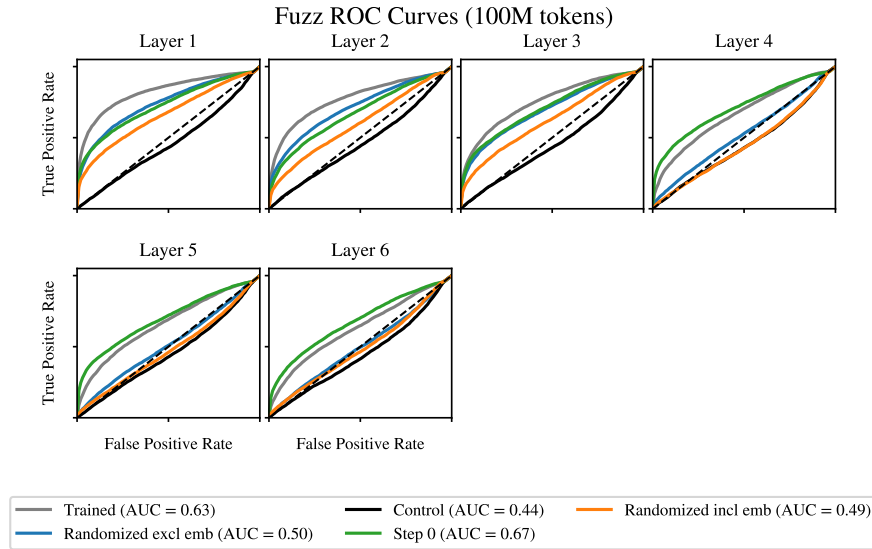


Figure 6: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-70m, computed over 100 sampled SAE latents. The curves show similar performance across SAE variants and a general decrease in score as layer index increases.

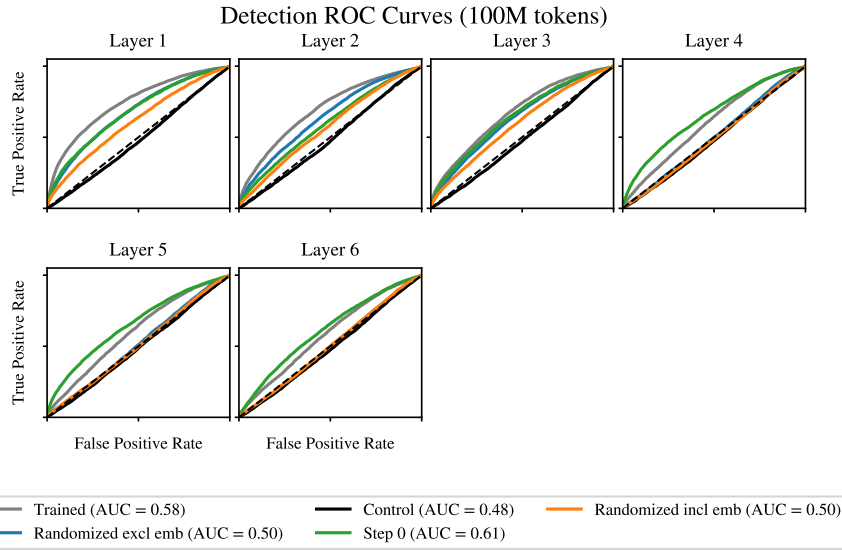


Figure 7: ‘Detection’ auto-interpretability ROC curves for Pythia-70m, computed over 100 sampled SAE latents. The curves show similar performance across SAE variants and a general decrease in score as layer index increases.

B.2 PYTHIA 160M

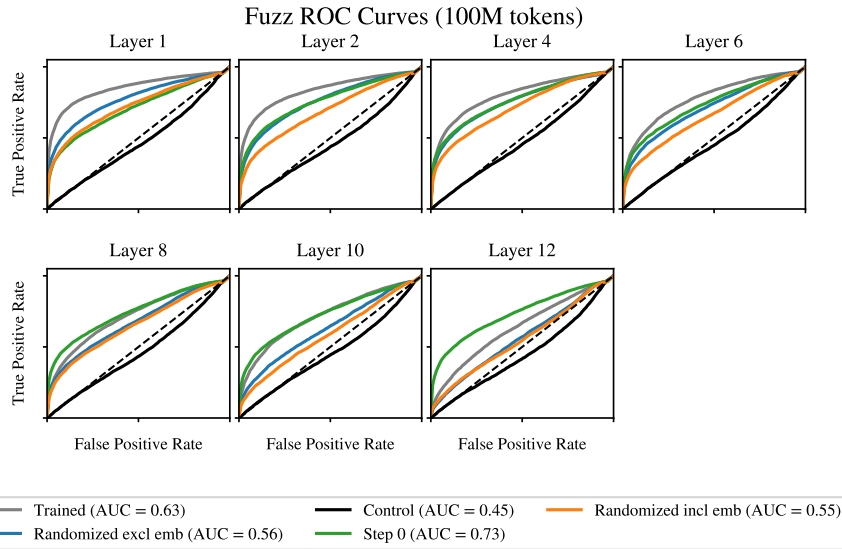


Figure 8: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-160m, computed over 100 sampled SAE latents. The variants exhibit similar performance profiles, with scores generally decreasing as layer index increases.

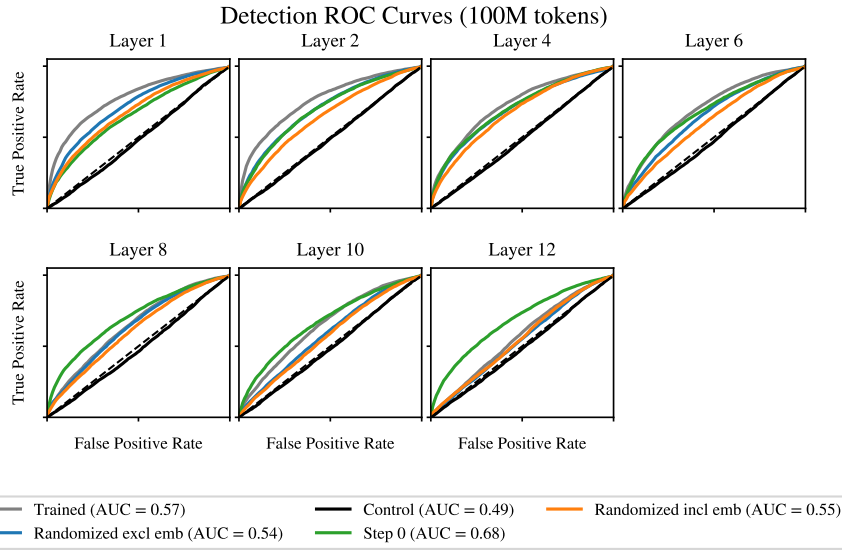


Figure 9: ‘Detection’ auto-interpretability ROC curves for Pythia-160m, computed over 100 sampled SAE latents. The variants exhibit similar performance profiles, with scores generally decreasing as layer index increases.

B.3 PYTHIA 410M

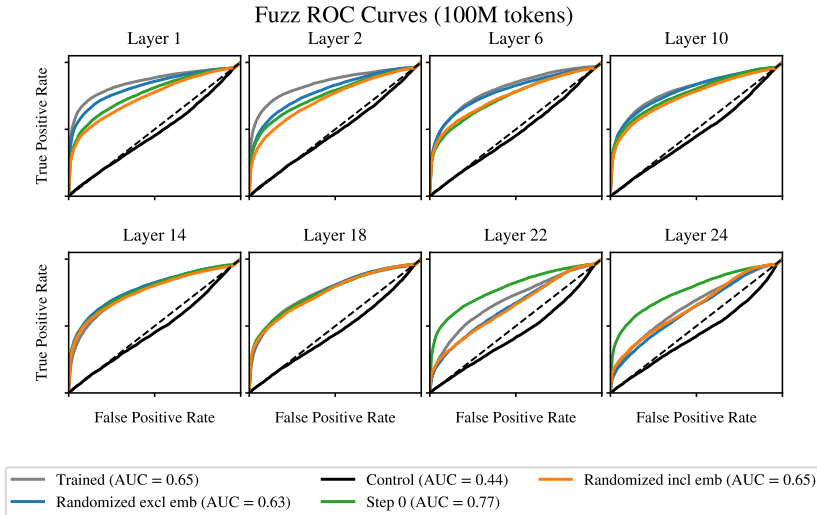


Figure 10: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-410m, computed over 100 sampled SAE latents. The variants exhibit similar performance profiles and a general score decrease with layer index; in this setting, the auto-interpretability scores do not separate trained and randomized models.

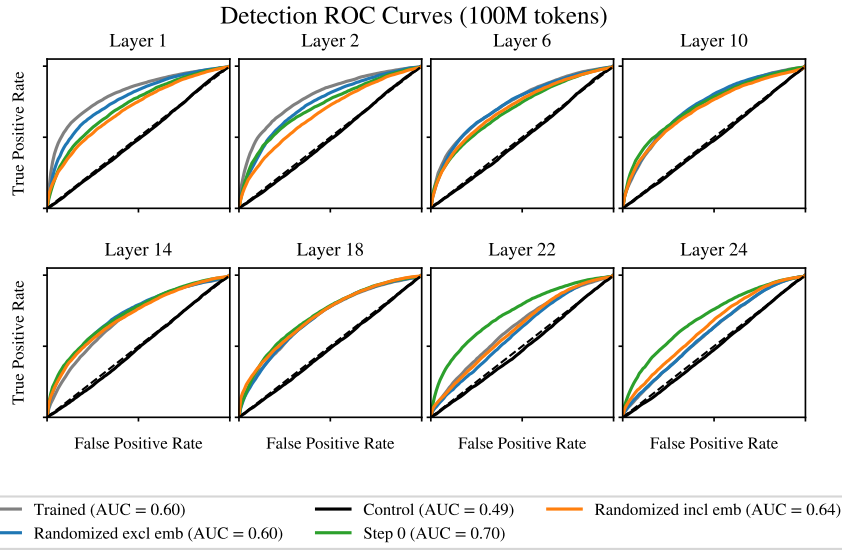


Figure 11: ‘Detection’ auto-interpretability ROC curves for Pythia-410m, computed over 100 sampled SAE latents. The variants exhibit similar performance profiles, with scores generally decreasing as layer index increases.

B.4 PYTHIA-1B

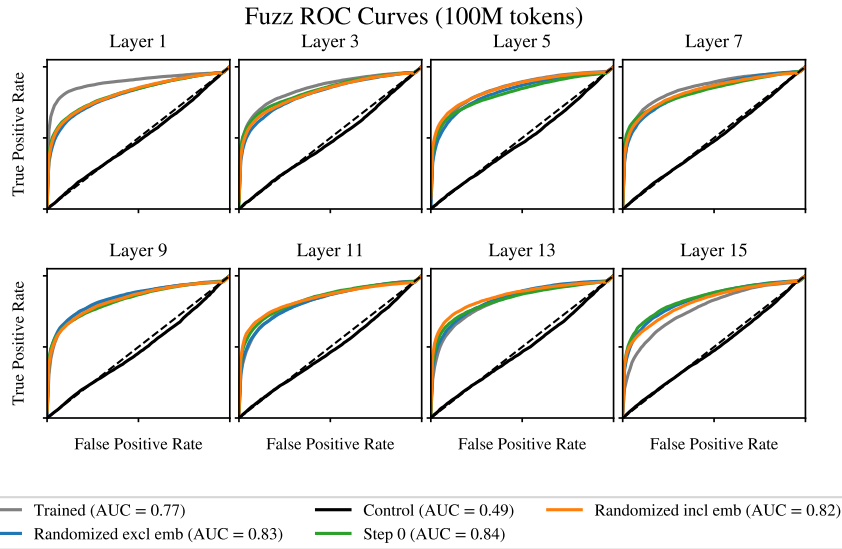


Figure 12: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-1b, computed over 100 sampled SAE latents. The variants show similar performance, but unlike the smaller models, the curves do not show a consistent degradation in quality across layers.

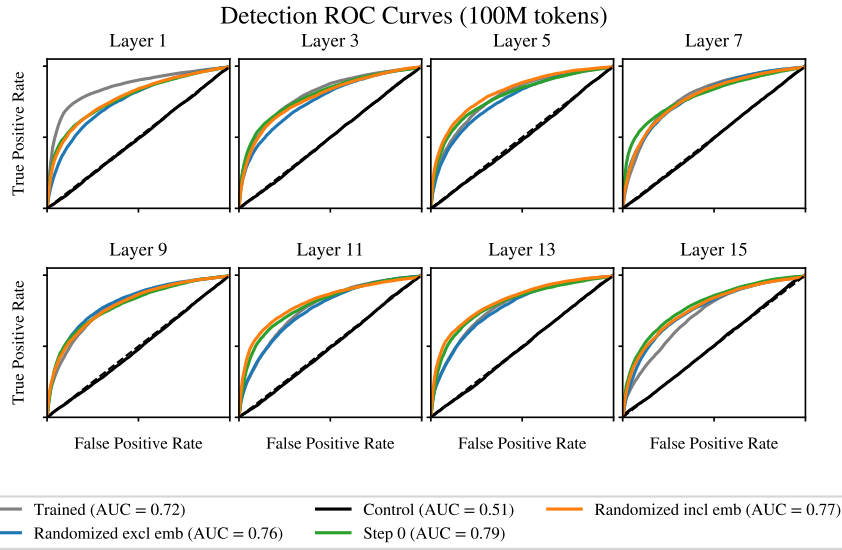


Figure 13: ‘Detection’ auto-interpretability ROC curves for Pythia-1b, computed over 100 sampled SAE latents. The variants show similar performance, but unlike the smaller models, the curves do not show a consistent degradation in quality across layers.

B.5 PYTHIA 6.9B

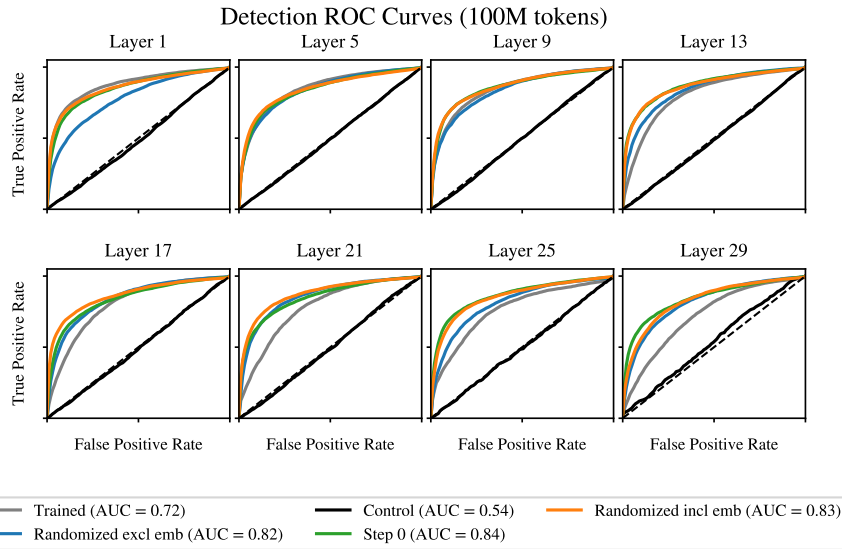


Figure 14: ‘Detection’ auto-interpretability ROC curves for Pythia-6.9b, computed over 100 sampled SAE latents. The variants obtain similar performance under the detection protocol.

C EFFECT OF INCREASED TRAINING DATA

The primary experiments train SAEs on 100M tokens (Section 3). To test whether the observed patterns are explained by insufficient SAE training data, we repeated a subset of experiments using SAEs trained on 1B RedPajama tokens (Figure 15).

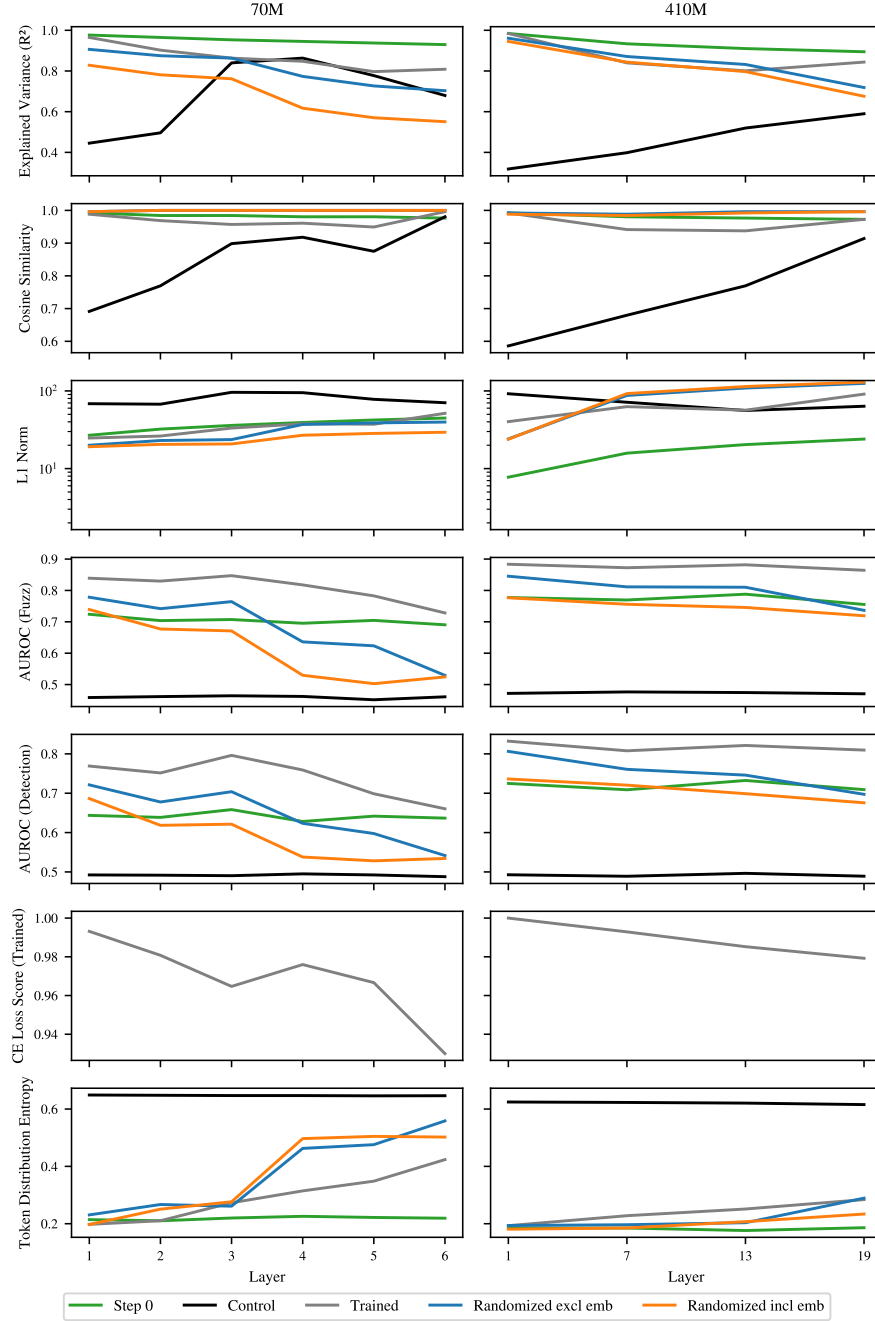


Figure 15: Evaluation metrics for SAEs trained on one billion tokens for the Pythia-70m and Pythia-410m models. These panels correspond to the same metric columns in Figure 2, where SAEs are trained on 100M tokens, and they exhibit qualitatively similar behavior.

D EFFECT OF DECREASED TRAINING DATA FOR PYTHIA-1B

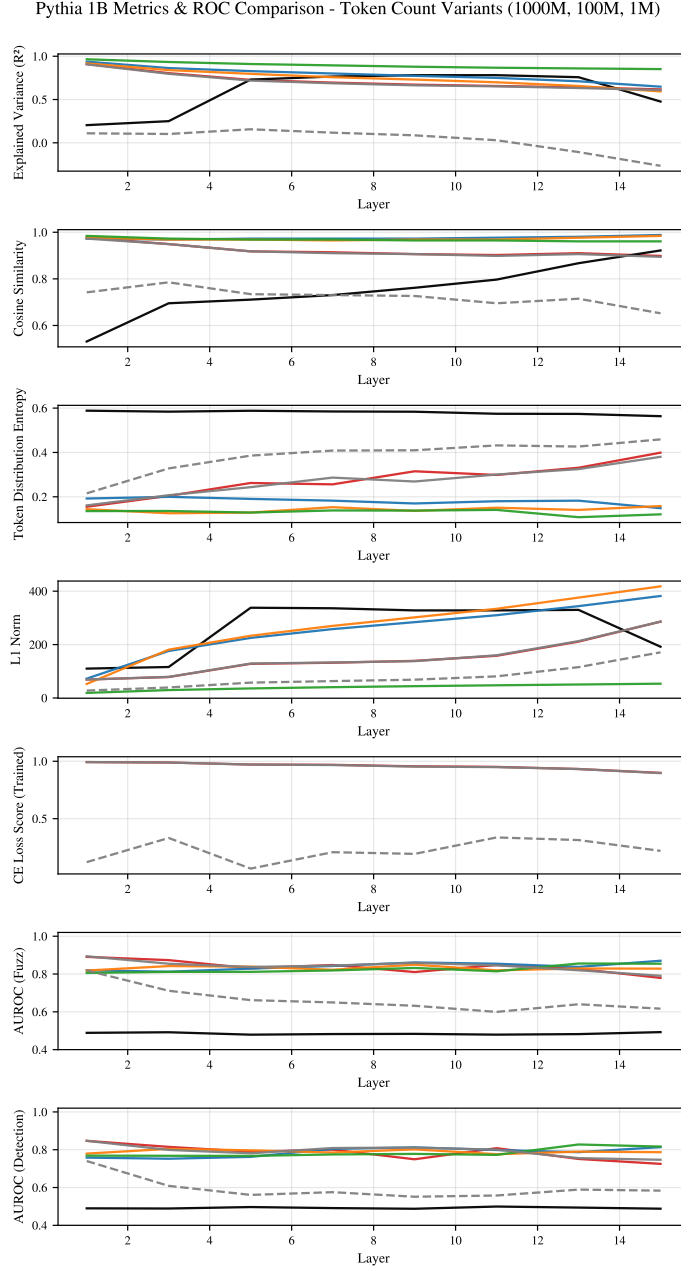
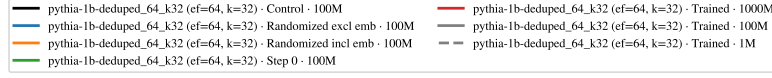


Figure 16: Evaluation metrics for SAEs trained on Pythia-1b using 1M and 1B tokens. The 1M-token SAEs have substantially lower explained variance and CE loss score, indicating under-training. Average auto-interpretability is slightly lower in the earliest layers and then declines sharply with layer index, while the layerwise trends in auto-interpretability and token-distribution entropy remain consistent with the other SAE runs.

E UNCERTAINTY PLOTS FOR PYTHIA-70M

We estimated uncertainty in the Pythia-70m evaluation metrics by repeating the experiment with five random seeds.

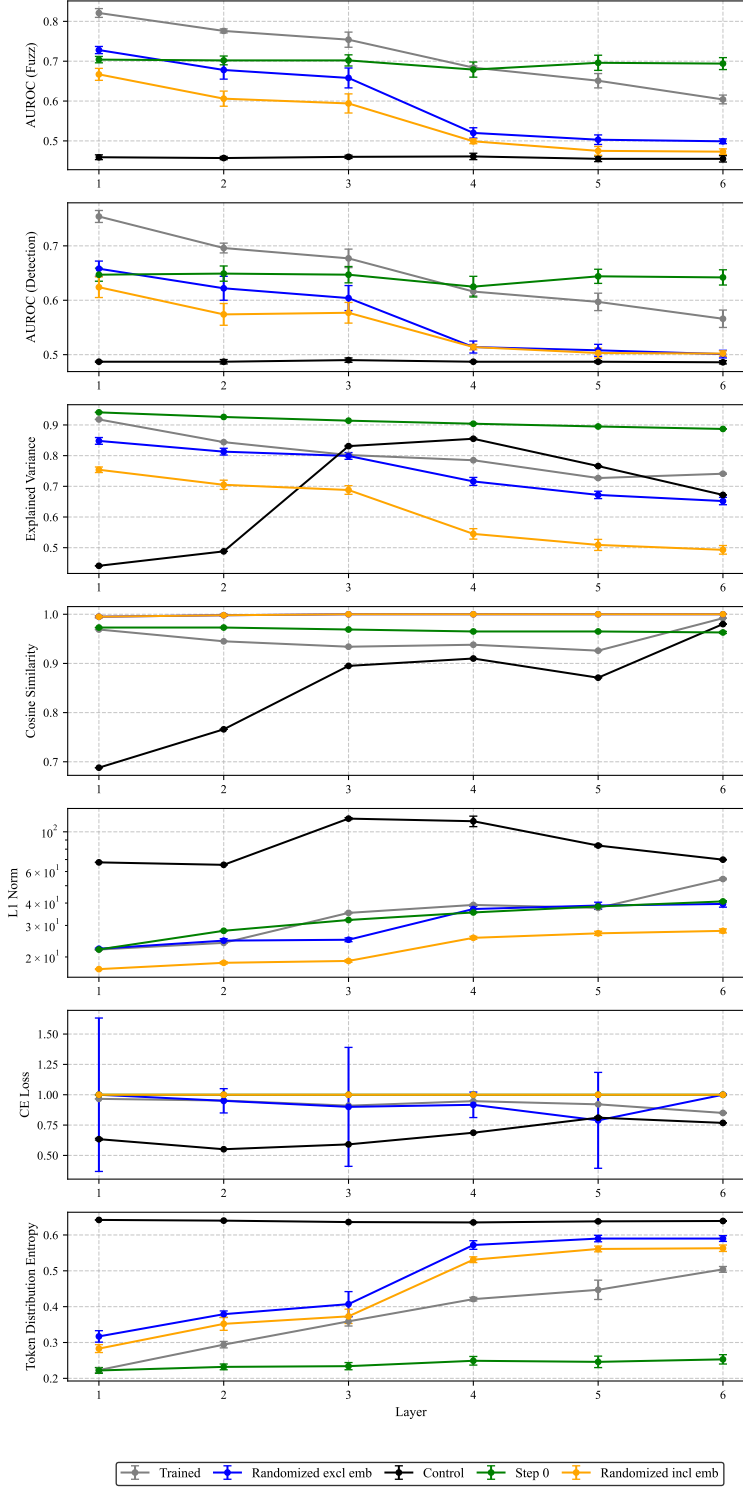


Figure 17: Uncertainty estimates for Pythia-70m metrics computed from five random seeds.

F EFFECT OF SAE HYPERPARAMETERS FOR PYTHIA-160M

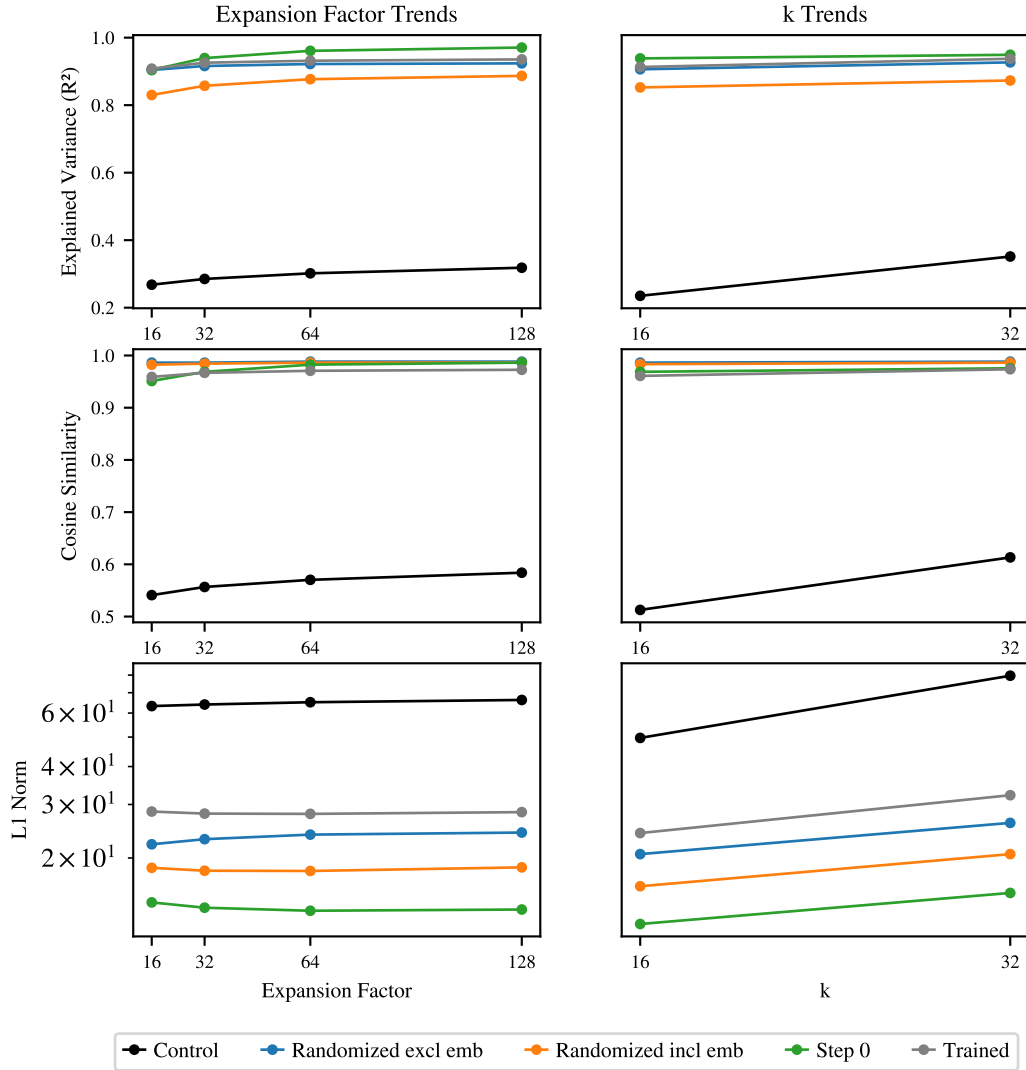


Figure 18: Robustness of SAE evaluation metrics to hyperparameter selection. Standard metrics remain stable across expansion factors R from 16 to 128 and sparsities k from 16 to 32, while initialization strategies preserve their relative ordering. This stability indicates that moderate default values, such as expansion factor $R = 64$ and sparsity $k = 32$, are sufficient for the comparisons reported here.

G EFFECT OF SAE HYPERPARAMETERS FOR PYTHIA-1B

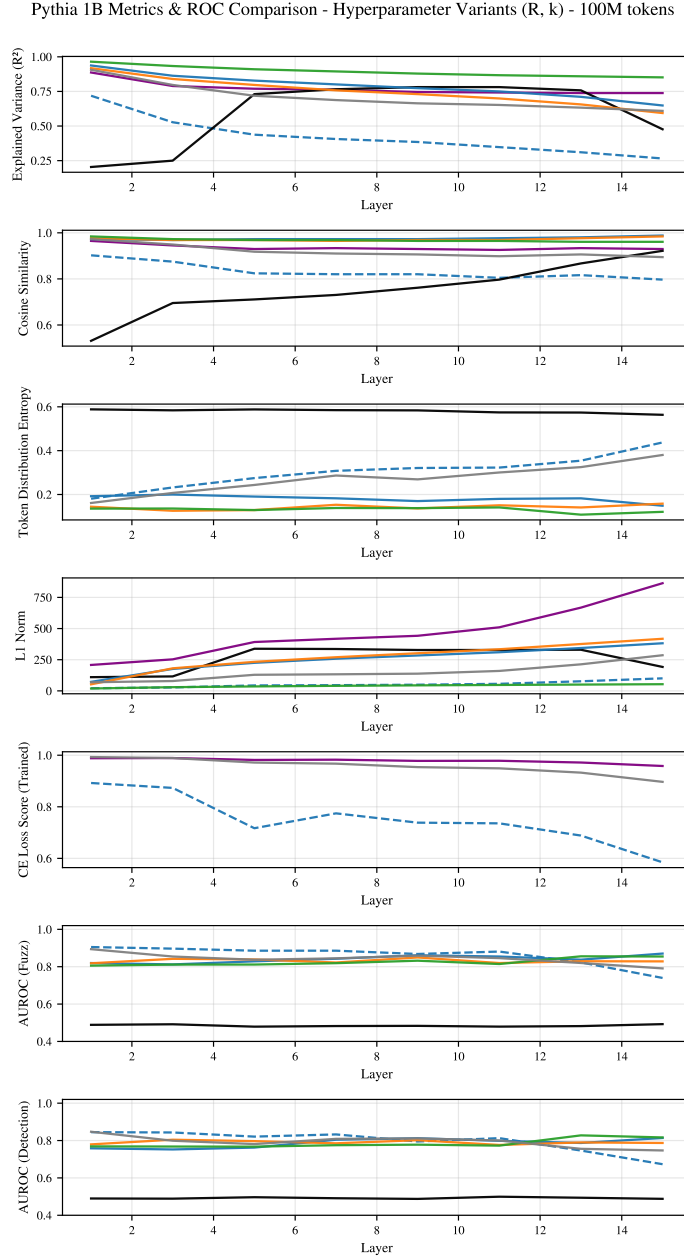


Figure 19: Evaluation metrics for SAEs trained on Pythia-1b with alternative hyperparameters, including the default setting from Figure 2. SAEs with expansion factor $R = 2$ and sparsity $k = 4$ are separated from the default setting by explained variance and CE loss score. However, their auto-interpretability scores remain similar to those of SAEs trained with default hyperparameters on both trained and randomised models.

H TOKEN DISTRIBUTION ENTROPY VS. AUTO-INTERPRETABILITY

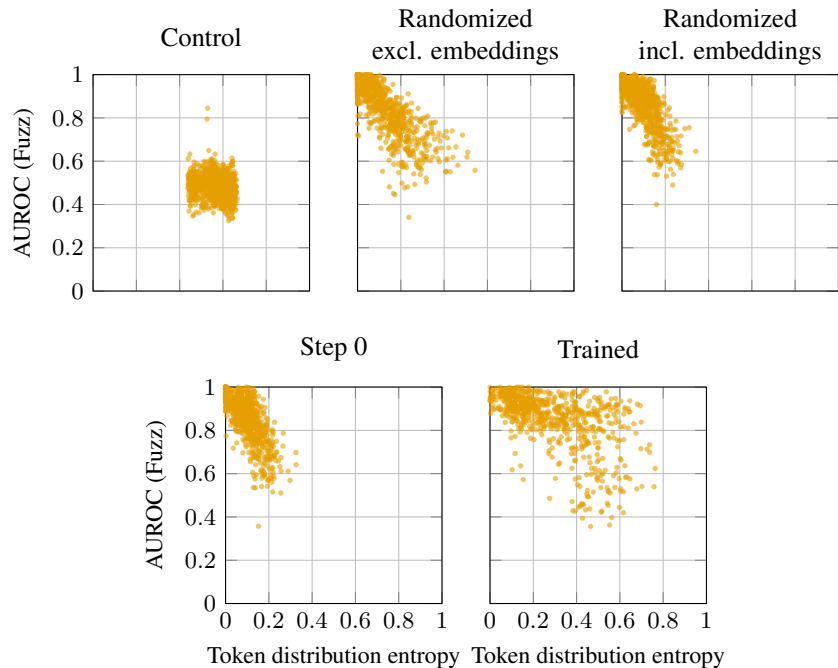


Figure 20: Scatter plots of per-latent token-distribution entropy against ‘fuzzing’ AUROC (auto-interpretability score) for SAEs trained on multiple Pythia-6.9b layers. Each point is one latent from the sample used to compute the aggregate metrics in Figures 1 and 2.

Figure 20 separates the negative control, randomized variants, and trained variant along entropy and auto-interpretability dimensions:

- **Control:** Latents have consistently high entropy, meaning that maximally activating examples distribute activation across many token IDs, and low auto-interpretability, meaning that generated explanations do not adequately predict the activation patterns. No clear correlation between the two variables is visible.
- **Randomized:** For each randomization scheme described in Section 3, entropy and auto-interpretability are negatively correlated: as a latent activates on a broader set of token IDs, its generated explanation less accurately accounts for the activation pattern.
- **Trained:** The correlation is weaker. In addition to the broad trend observed for randomized variants, some latents have both high entropy and high auto-interpretability, indicating activation patterns distributed across multiple token IDs that remain consistent with the generated explanation.

These results support the interpretation that aggregate auto-interpretability scores obscure differences between SAEs trained on trained and randomized backbones. Randomized models with stable token embeddings can produce easy-to-explain ‘single-token’ features, whereas transformers trained on natural language additionally produce more complex semantic features.

I A TOY MODEL OF SUPERPOSITION

Section 4 trains SAEs on synthetic data constructed to exhibit superposition (Sharkey et al., 2022) and on GloVe word vectors (Pennington et al., 2014). This appendix specifies the data-generation procedure and training configuration used for those experiments.

I.1 DATA GENERATION

We first define the ground-truth feature dictionary by sampling n_s points on an n_d -dimensional hypersphere.

For each sample, feature coefficients are generated by drawing $A \in \mathbb{R}^{n_s \times n_s}$ with entries $A_{ij} \sim \mathcal{N}(0, 1)$, constructing the covariance matrix $\Sigma = AA^\top$, sampling $\vec{\alpha} \in \mathbb{R}^{n_s}$ where $\alpha_i \sim \mathcal{N}(\vec{0}, \Sigma)$, mapping each α_i through the c.d.f. of $\mathcal{N}(0, 1)$, applying the decay transformation $\alpha_i \rightarrow \alpha_i^{\lambda}$ where $\lambda \in \mathbb{R}$, normalizing via $\alpha_i \rightarrow m\alpha_i / n_s \sum_j \alpha_j$ where $m \in \mathbb{R}$, and then performing n_s independent Bernoulli trials with $p = \alpha_i$. The Bernoulli outcomes are then multiplied by n_s independent samples from the continuous uniform distribution $\mathcal{U}_{[0,1]}$.

The parameter λ controls the decay rate of nonzero ground-truth feature frequency as a function of feature index i . The parameter m specifies the expected number of nonzero feature coefficients per sample.

Following Sharkey et al. (2022), we set $n_s = 512$, $n_d = 256$, $\lambda = 0.99$, and $m = 5$. Figure 21 provides a Python implementation of the procedure.

I.2 TRAINING

The SAEs used in Section 4 consist of a linear encoder with bias, a ReLU nonlinearity, and a linear decoder without bias. Decoder weights are initialized orthogonally, and decoder weight vectors are normalized before every training step.

The training objective is the mean squared error (MSE) between input and reconstruction plus a coefficient-weighted mean L^1 norm of the encoded representation; the coefficient is varied from 1×10^{-3} to 100.

For the synthetic data, models are trained for 100 epochs on 10K samples across 10 random seeds. For word-vector data, models are trained for 100 epochs on 400K samples with one random seed. In both settings, 10% of samples are held out as a validation set for metric computation.

The MLPs used in Section 4 have two linear layers, equivalently one hidden layer, with a ReLU activation. Both input and output dimensionality are n_d , and the hidden dimensionality is $4n_d$. This configuration is loosely motivated by the feed-forward sublayers of transformer language models.

```

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1414
1415
1416 1 def generate_sharkey(
1417 2     num_samples: int,
1418 3     num_inputs: int,
1419 4     num_features: int,
1420 5     avg_active_features: float,
1421 6     lambda_decay: float,
1422 7 ) -> tuple[Tensor, Tensor]:
1423 8     """
1424 9     Args:
1425 10         num_samples (int): The number of samples to generate.
1426 11         num_inputs (int): The number of input dimensions.
1427 12         num_features (int): The number of ground truth features.
1428 13         avg_active_features (float): The average number of
1429 14             ground truth features active at a time.
1430 15         lambda_decay (float): The exponential decay factor for
1431 16             feature probabilities.
1432 17     """
1433 18     features = torch.randn(num_inputs, num_features)
1434 19     features /= torch.norm(features, dim=0, keepdim=True)
1435 20
1436 21     covariance = torch.randn(num_features, num_features)
1437 22     covariance = covariance @ covariance.T
1438 23     correlated_normal = MultivariateNormal(
1439 24         torch.zeros(num_features), covariance_matrix=covariance
1440 25     )
1441 26
1442 27     samples = []
1443 28     for _ in range(num_samples):
1444 29         p = STANDARD_NORMAL.cdf(correlated_normal.sample())
1445 30         p = p ** (lambda_decay * torch.arange(num_features))
1446 31         p = p * (avg_active_features / (num_features * p.mean()))
1447 32         p = torch.bernoulli(p.clamp(0, 1))
1448 33         coef = p * torch.rand(num_features)
1449 34
1450 35         sample = coef @ features.T
1451 36         samples.append(sample)
1452 37
1453 38     return torch.stack(samples), features

```

Figure 21: Python implementation of the data-generation procedure from Sharkey et al. (2022) used in Section 4.

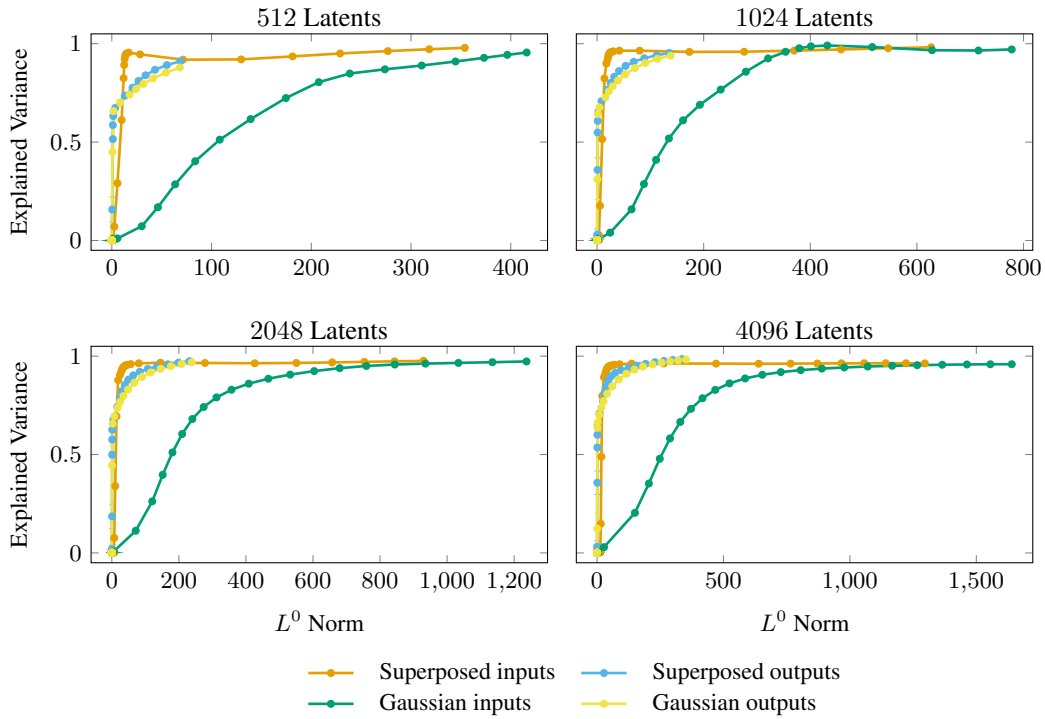


Figure 22: Pareto frontiers of explained variance against the L^0 norm (sparsity) for toy datasets constructed to exhibit superposition, Gaussian controls matched in mean and variance, and the corresponding outputs produced by a randomly initialized two-layer MLP.

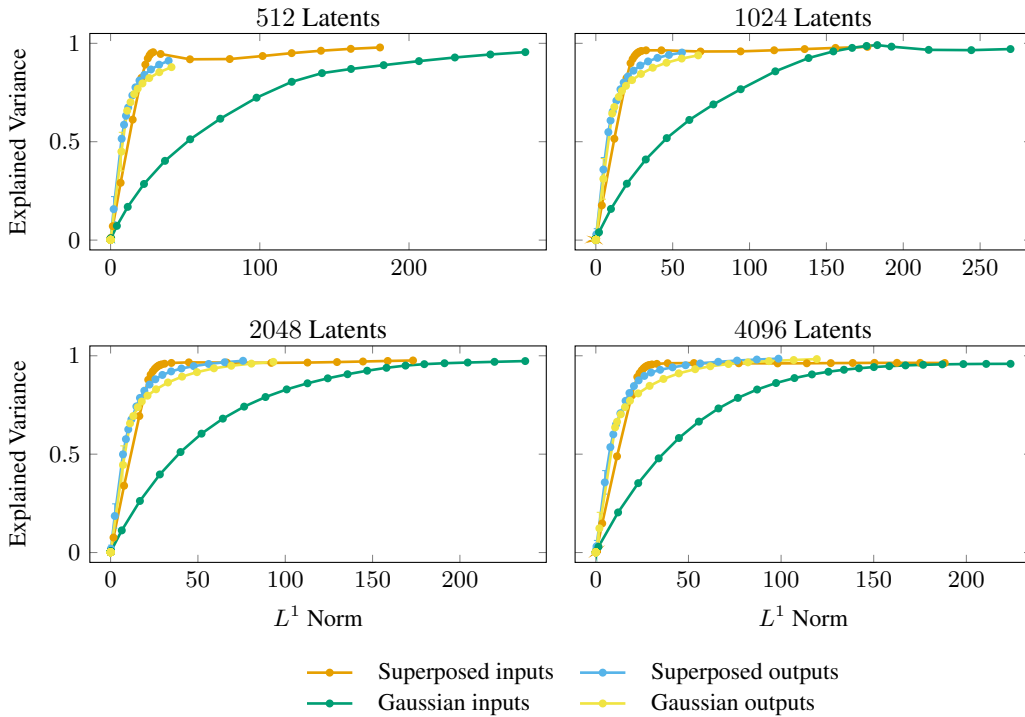


Figure 23: Pareto frontiers of explained variance against the L^1 norm (sparsity) for toy datasets constructed to exhibit superposition, Gaussian controls matched in mean and variance, and the corresponding outputs produced by a randomly initialized two-layer MLP.

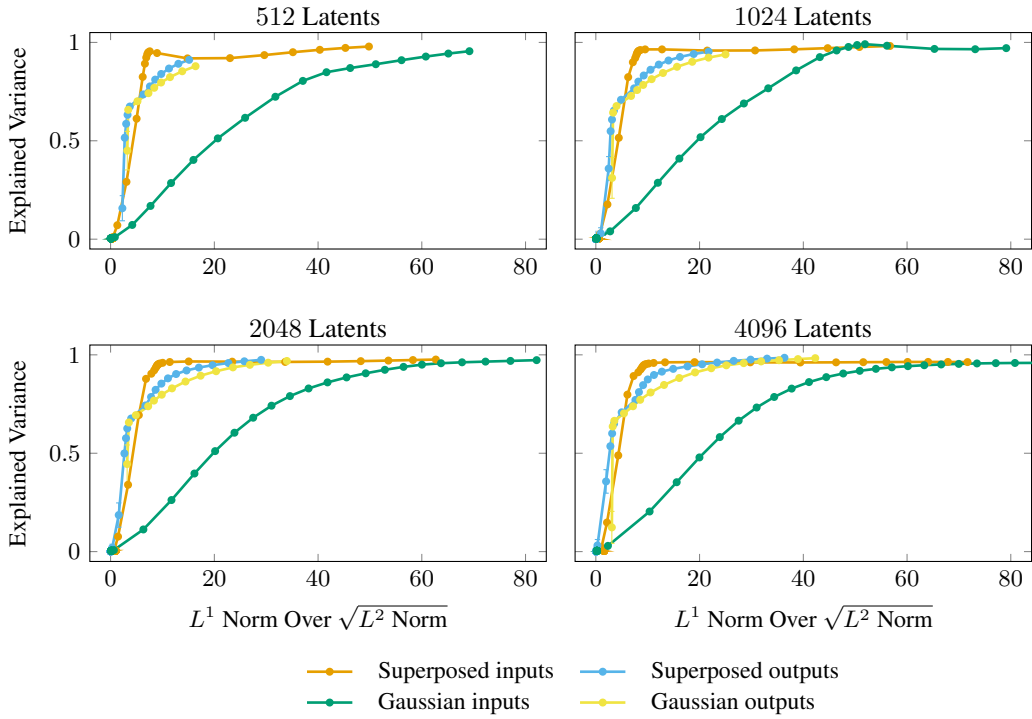


Figure 24: Pareto frontiers of explained variance against the L^1 norm divided by the square root of the L^2 norm (sparsity) for toy datasets constructed to exhibit superposition, Gaussian controls matched in mean and variance, and the corresponding outputs produced by a randomly initialized two-layer MLP.

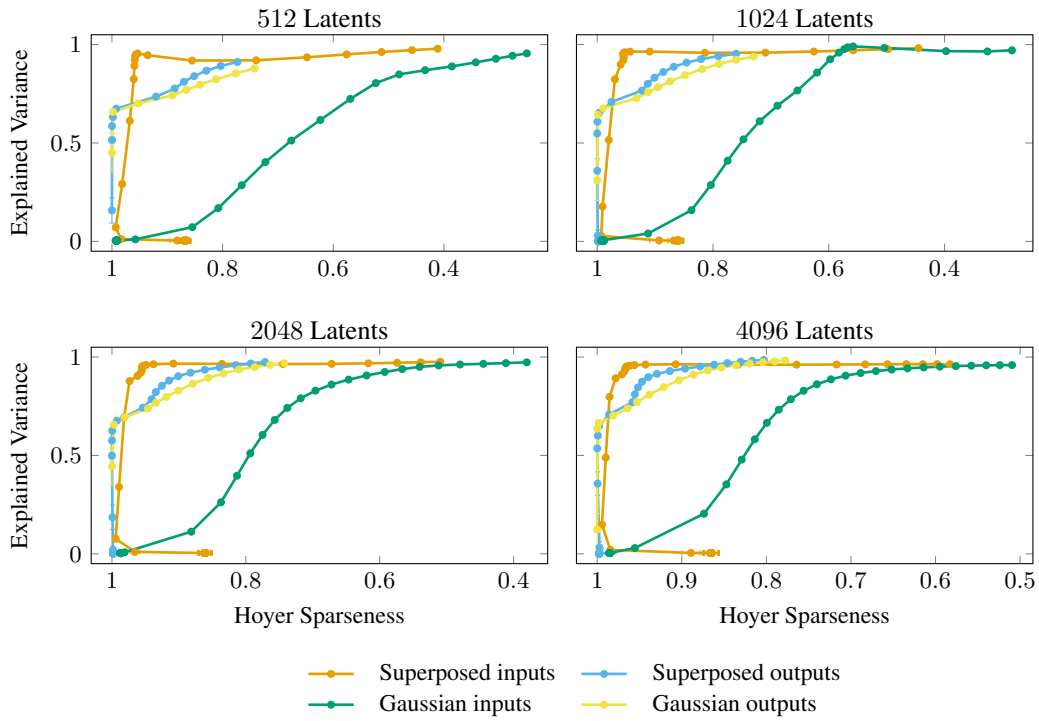


Figure 25: Pareto frontiers of explained variance against Hoyer sparseness for toy datasets constructed to exhibit superposition, Gaussian controls matched in mean and variance, and the corresponding outputs produced by a randomly initialized two-layer MLP.

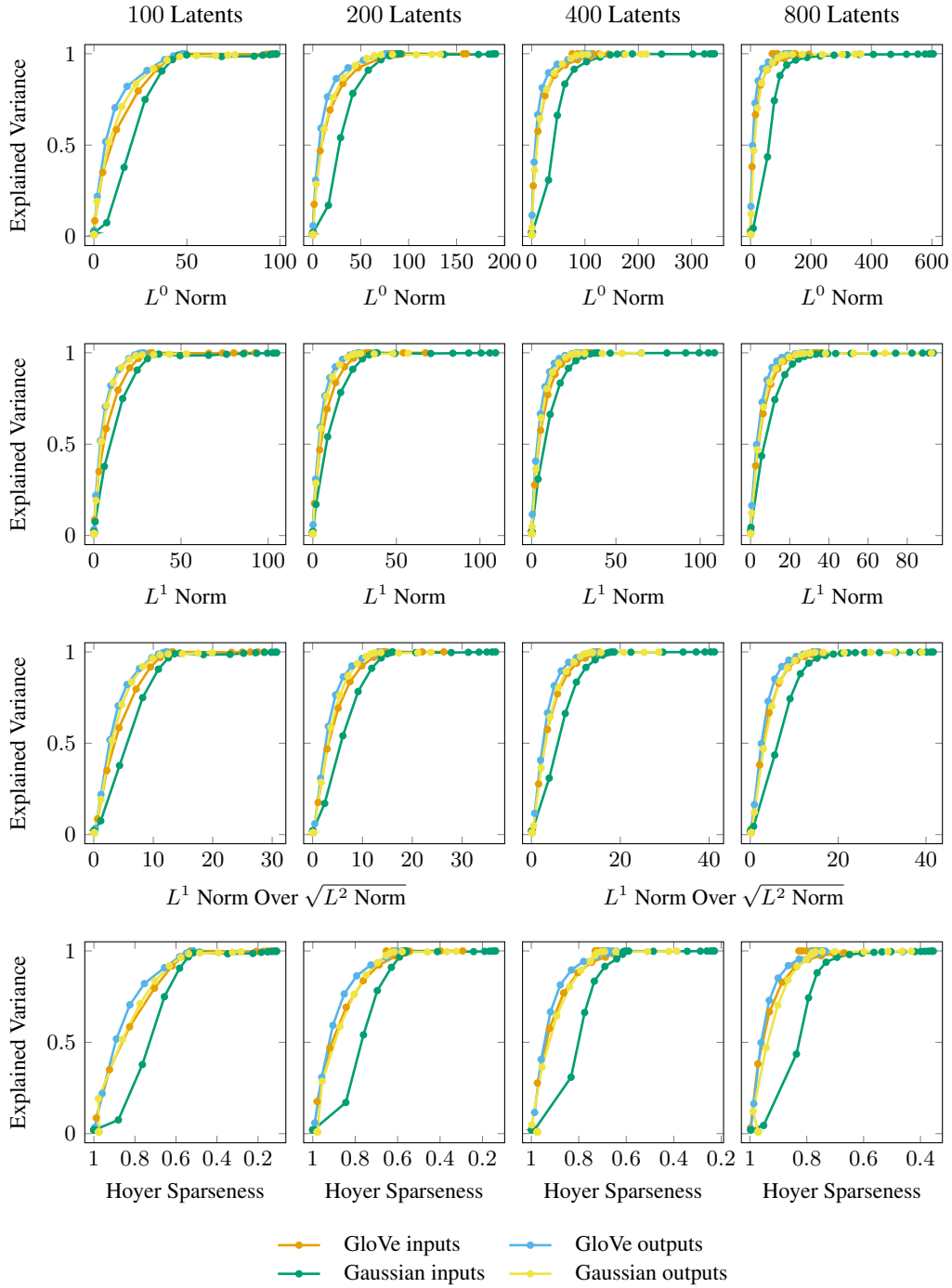


Figure 26: Pareto frontiers of explained variance against sparsity measures for 50-dimensional GloVe word vectors, Gaussian controls matched in mean and variance, and the corresponding outputs produced by a randomly initialized two-layer MLP.

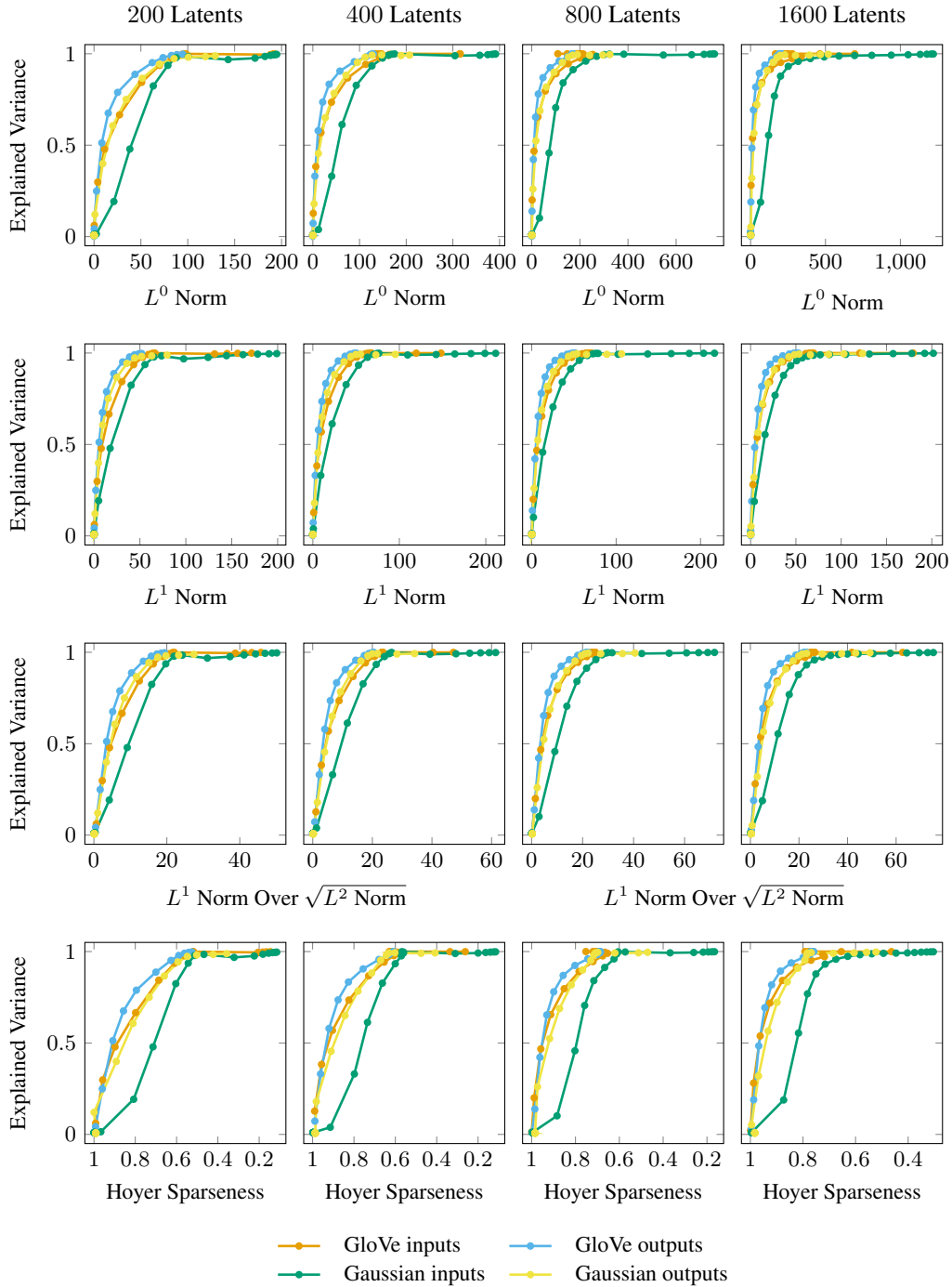


Figure 27: Pareto frontiers of explained variance against sparsity measures for 100-dimensional GloVe word embeddings, Gaussian controls matched in mean and variance, and the corresponding outputs produced by a randomly initialized two-layer MLP.

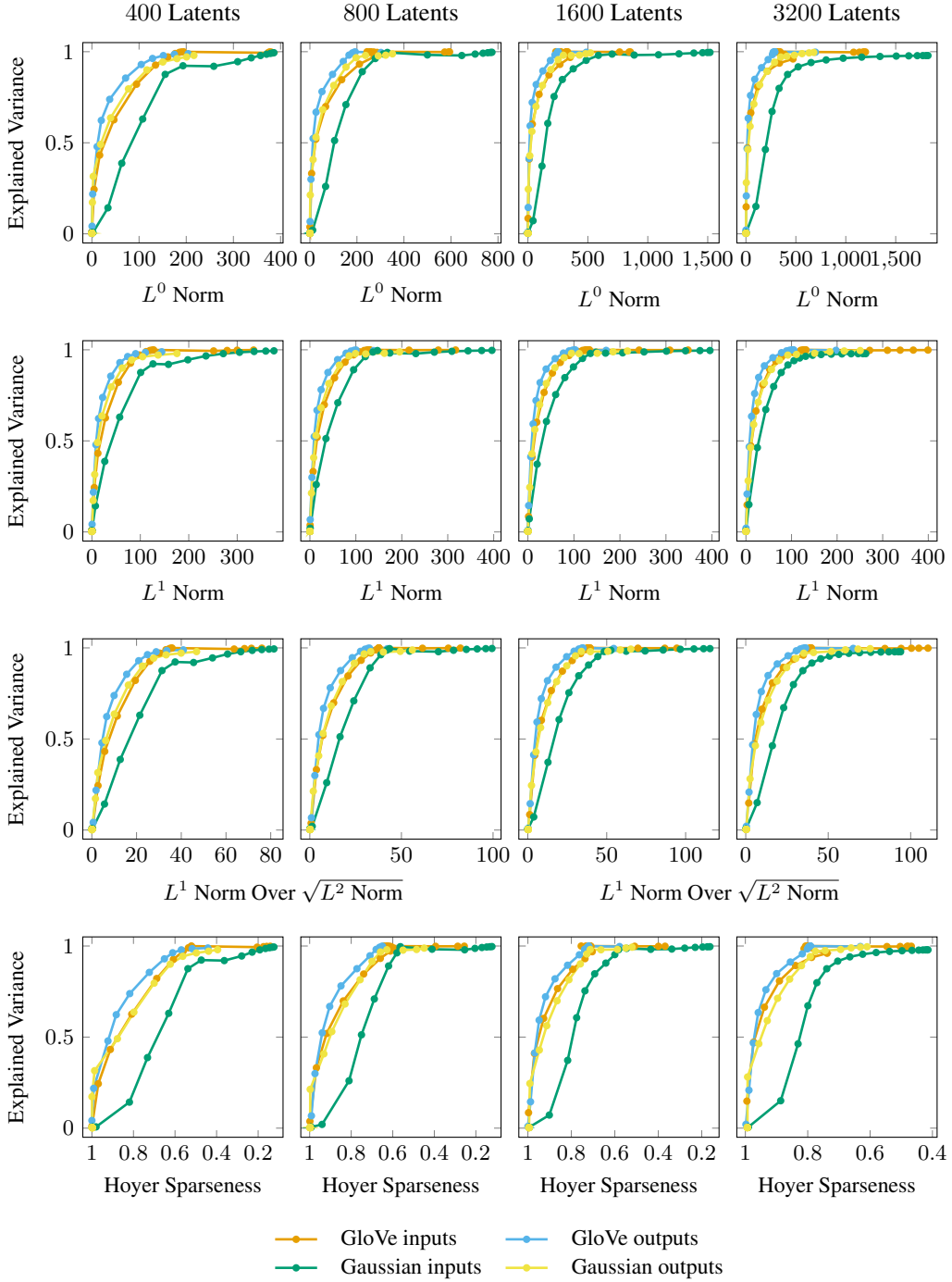


Figure 28: Pareto frontiers of explained variance against sparsity measures for 200-dimensional GloVe word embeddings, Gaussian controls matched in mean and variance, and the corresponding outputs produced by a randomly initialized two-layer MLP.

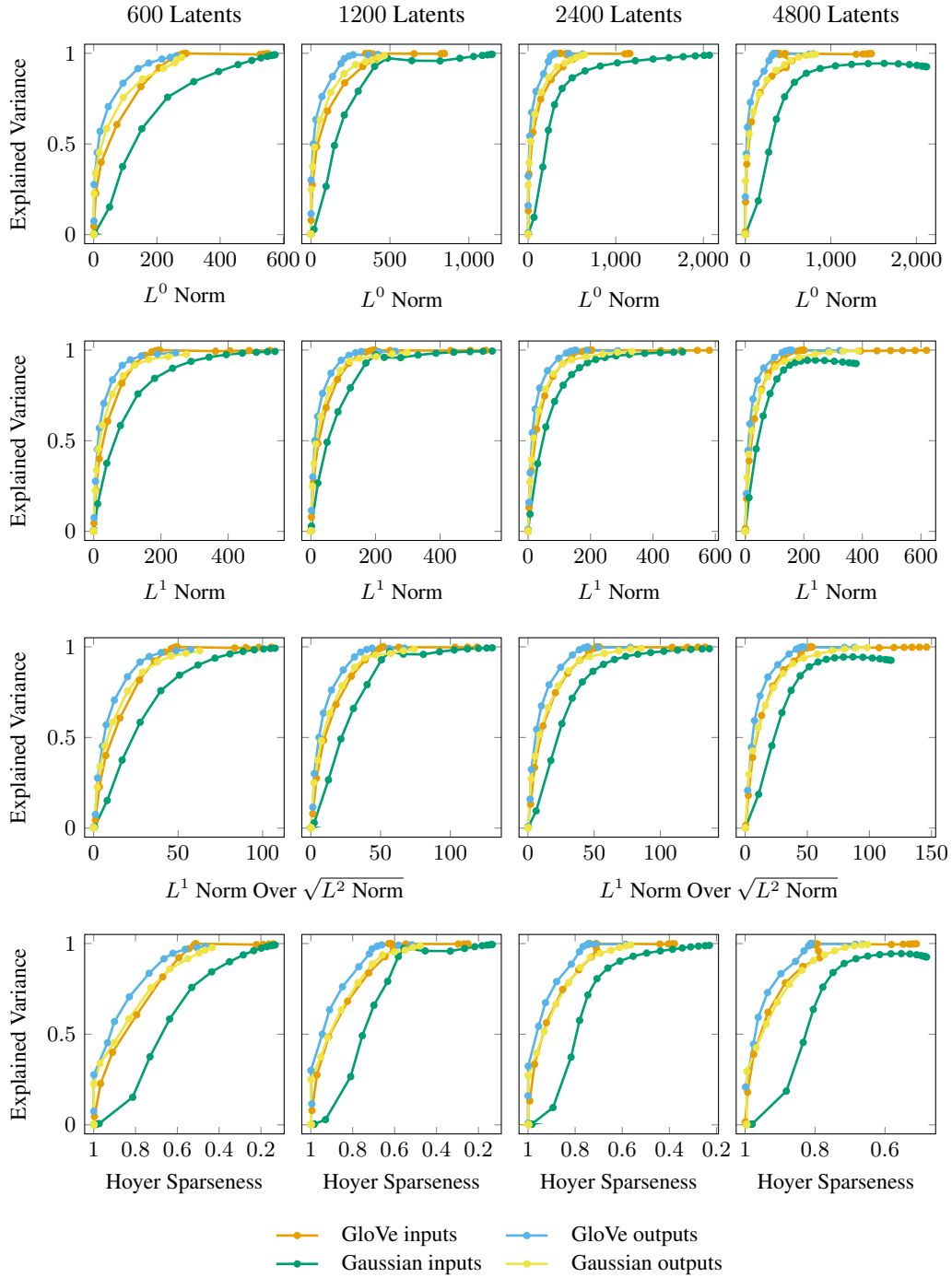


Figure 29: Pareto frontiers of explained variance against sparsity measures for 300-dimensional GloVe word embeddings, Gaussian controls matched in mean and variance, and the corresponding outputs produced by a randomly initialized two-layer MLP.

1836 J EXAMPLE FEATURES FOR PYTHIA-6.9B

1837
1838 VARIANT: TRAINED

1839
1840 FEATURE 180935 (LAYER 0)

1841
1842 **Interpretation:** The term "security" is predominantly used to refer to protection, safety, and measures
1843 to prevent harm, while "oz" is likely referring to ounces, possibly in a context of measurement or
1844 quantification, although "oz" appears less frequently and often in a different context.

1845 **Top Examples:**

- 1846
1847 Text: <endoftext>—ζ—. If for any reason you are unhappy with our service please contact us
1848 directly so we can make it right for you. Journal of Cyber Security, Vol.
1849 1. Activation: 4.3750
1850 Active tokens: Security
1851 Text: <endoftext>—ζ— Security Practitioner. Tremendously passing CompTIA Advanced
1852 Security Practitioner (casp) cert has never been as easy as it
1853 2. Activation: 4.3125
1854 Active tokens: Security Security
1855 Text: in trusted hands for your Cyber Security career or staffing needs. Call 0203 643 0248 to
1856 find out more. Technically proficient using
1857 3. Activation: 4.3125
1858 Active tokens: Security
-

1859
1860 FEATURE 93790 (LAYER 8)

1861
1862 **Interpretation:** Nouns and phrases related to economic concepts, development, and business, often
1863 referring to growth, progress, and improvement.

1864 **Top Examples:**

- 1865
1866 Text: training requirements. See "Workforce" section for additional information. The Economic
1867 Development Transportation Fund, commonly referred to as the "Road Fund," is an
1868 1. Activation: 21.3750
1869 Active tokens: Development
1870 Text: Montréal. The Williamsburg Economic Development Authority offers a 33% matching
1871 grant up to \$7,500 for exterior improvements to existing businesses in the City of
1872 2. Activation: 20.2500
1873 Active tokens: Development Authority
1874 Text: Correction: In a July 16 web story The Real Deal incorrectly stated that the Economic
1875 Development Corporation was "circumventing" laws with its restructuring. In
1876 3. Activation: 20.0000
1877 Active tokens: Development
-

1878
1879
1880 FEATURE 128309 (LAYER 12)

1881
1882 **Interpretation:** Various types of punctuation and grammatical elements that separate words or
1883 phrases, including hyphens, commas, ellipses, prepositions, and determiners, often indicating connec-
1884 tions, contrasts, or clarifications, and sometimes marking boundaries or transitions between clauses
or ideas.

1885 **Top Examples:**

- 1886
1887 Text: Run it in JDK6, and it will print "[axons, bandrils, chumbliēs]". If you are having trouble
1888 switching from
1889 1. Activation: 8.0000
Active tokens: in JD K

-
- 1890 Text: Here, we introduce the coordinate systems for three-dimensional space □□□2. The study
1891 of 3-dimensional spaces lead us to the setting for our study
1892 2. Activation: 7.8125
1893 Active tokens: □
1894 Text: .path.expanduser("~/malwarehouse/") because this server doesn't have X-Windows running.
1895 If you are looking for a simple and
1896 3. Activation: 7.7188
1897 Active tokens: . path expand user
-

1898
1899
1900
1901 VARIANT: STEP 0
1902

1903 FEATURE 126848 (LAYER 12)
1904

1905 **Interpretation:** Nouns denoting people who train others, units or marks of measurement, and
1906 abbreviations or acronyms representing specific standards or technologies.
1907

1908 **Top Examples:**

- 1909 Text: What are the various lessons a member can access at a tennis club? Whether you are a
1910 1. beginner or advanced player, trainers help you to choose the right gaming
1911 Activation: 13.1250
1912 Active tokens: trainers
1913 Text: report include various simulation platforms and Serious Games. The report also analyzes
1914 some major allied products such as patient simulators and task trainers. The technologies
1915 2. analyzed
1916 Activation: 13.0625
1917 Active tokens: trainers
1918 Text: a stylish spring in your step when you buy from our fantastic range of men's and women's
1919 Asics trainers. We've got numerous styles from
1920 3. Activation: 13.0000
1921 Active tokens: trainers
-

1922
1923
1924 FEATURE 2125 (LAYER 4)
1925

1926 **Interpretation:** The word "papers" is often used in contexts referring to written documents, such as
1927 academic papers, court documents, or printed materials, and is frequently mentioned in relation to
1928 tasks like writing, research, and education.
1929

1930 **Top Examples:**

- 1931 Text: caustic solution . As an abrasive, alumina is coated into abrasive papers and .. Pakistan.
1932 Sierra leone. Taiwan. Turkey. Venezuela.
1933 1. Activation: 5.7188
1934 Active tokens: papers
1935 Text: that can be associated with interaction with other individuals. For everybody who is
1936 uncertain regardless of whether your papers is misstep no cost, buy inexpensive experienced
1937 2. proofreading services
1938 Activation: 5.6875
1939 Active tokens: papers
1940 Text: who RV, often traveling in groups, often alone. You just want to have all the papers like
1941 RC, licence and insurance coverage as effectively as PUC (
1942 3. Activation: 5.6562
1943 Active tokens: papers
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1997

FEATURE 9944 (LAYER 16)

Interpretation: Words or parts of words that are usually the beginning or end of a proper noun, surname, or a word of foreign origin.

Top Examples:

1. Text: astic gestures.Home Music World News Music the artform Where Do Music Festivals Go Now? Where Do Music Festivals Go Now? Are you ready
Activation: 10.4375
Active tokens: Fest Fest
 2. Text: client streams, and encouraging existing customers to become more involved.Welcome to Fil Fest USA!! Do you love lumpia? Can you eat a handful of them
Activation: 10.3750
Active tokens: Fest
 3. Text: Powers talking about his early inspirations. In response to San Diego Comic Fest 2012!: Off to Comic Fest 2012. Should be interesting if nothing else!
Activation: 10.3750
Active tokens: Fest Fest
-

VARIANT: RANDOMIZED EXCLUDING EMBEDDINGS

FEATURE 151030 (LAYER 28)

Interpretation: Common nouns, proper nouns, or adjectives found in various contexts, including but not limited to geographical locations, people, organizations, time, and concepts, often possessing relevance to the surrounding text.

Top Examples:

1. Text: ion Patch Kills Owner, Son,”, Los Angeles Times, June 12, 1994, <http://articles.latimes.com/1994-06-12>
Activation: 58.0000
Active tokens: lat
 2. Text: hard-headed coin of the realm their look. Firstly you’ve got to inventory on incident unique a distinct blunt in a retailer you superlativeness be
Activation: 56.5000
Active tokens: lat
 3. Text: Jr’s new film Dovlatov follows the life of the now celebrated writer Sergei Dovlatov over six days in 1971, as he struggles
Activation: 55.7500
Active tokens: lat lat
-

FEATURE 98924 (LAYER 12)

Interpretation: Adjectives describing size, or nouns representing concepts or objects that are being described in terms of their size.

Top Examples:

1. Text: . The area to the right is for large dogs (small dogs also welcomed) however the area to the left is for small dogs only. Troup 69
Activation: 46.0000
Active tokens: small
2. Text: ,I would not mind, but has to pretty less expensive. Can it use any windows aplication???? Or I am really need a cool,small
Activation: 45.5000
Active tokens: small

1998 Text: 3/4” – 8 1/2” rather than strictly 8 1/4”) I chose the “small” version, though I should be a
1999 3. Activation: 44.7500
2000 Active tokens: small
2001

2002
2003 FEATURE 180589 (LAYER 24)
2004

2005 **Interpretation:** Nouns mostly referring to tasks, responsibilities or jobs to be accomplished, often in
2006 a professional or organizational context, sometimes accompanied by proper nouns and a few instances
2007 with words having suffixes or prefixes.
2008

2009 **Top Examples:**

- 2010 Text: completing tasks or a captcha, users are awarded by GRSfractions. Why These Groestlcoin
2011 1. Faucets provide rewards? Many people
2012 Activation: 130.0000
2013 Active tokens: tasks
2014 Text: from Groestlcoin Faucets is In the exchange of completing tasks or a captcha, users are
2015 2. awarded by Free GRS. To Earn
2016 Activation: 129.0000
2017 Active tokens: tasks
2018 Text: and still contain a small remnant circular genome, known as mitochondrial DNA. Of the
2019 varied tasks undertaken by mitochondria, the most important is the generation of the chemical
2020 3. energy
2021 Activation: 128.0000
2022 Active tokens: tasks
2023
-

2024
2025 VARIANT: RANDOMIZED INCLUDING EMBEDDINGS
2026

2027 FEATURE 39748 (LAYER 0)
2028

2029 **Interpretation:** Words contain “Pul” are often used in the context of Pulitzer, a prestigious journalism
2030 award, while “Looking” typically precedes a phrase expressing anticipation, expectation, or searching
2031 for something.
2032

2033 **Top Examples:**

- 2034 Text: <endoftext—¿— to the 21st Century. Rhodes won the Pulitzer prize for The Making of
2035 1. the Atomic Bomb (01987) his first of four books chronicling the
2036 Activation: 3.3750
2037 Active tokens: Pul
2038 Text: <endoftext—¿—, James Coburn, movies we love, Pulp Consumption, Steve McQueen,
2039 2. Western, Yul Brynner. Bookmark the permal
2040 Activation: 3.3438
2041 Active tokens: Pul
2042 Text: Crusher, Coal Mill and Coal Pulverizer for sale Coal crusher and coal mill is the major
2043 3. mining equipment in . sbm ceramic machinary -
2044 Activation: 3.2969
2045 Active tokens: Pul
2046
-

2047 FEATURE 15633 (LAYER 20)
2048

2049 **Interpretation:** Nouns representing individuals or entities possessing or having authority over
2050 something, often in a possessive or authoritative relationship with that thing.
2051

Top Examples:

-
- 2052 Text: poetry. The Starkville/Mississippi State University Symphony Orchestra kicks off 2012
2053 with a Jan. 21 concert dedicated to parents of the performing musicians. The free
2054 1. Activation: 80.0000
2055 Active tokens: Stark
- 2056 Text: Baptist. Kris Kirkwood, Stark Raving Solutions' lighting designer, says the architectural
2057 system used, ETC's Paradigm architectural control, is
2058 2. Activation: 77.5000
2059 Active tokens: Stark
- 2060 Text: so you can be as cool as Tony Stark. The Marvel Training Academy will be taking place
2061 throughout May, just check with your local shop to guarantee your place
2062 3. Activation: 77.5000
2063 Active tokens: Stark
-

2064
2065
2066 FEATURE 6069 (LAYER 4)

2068 **Interpretation:** Proper nouns, nouns referring to objects or places, and nouns with strong semantic
2069 connotations often related to religion or technology.

2070 **Top Examples:**

- 2071 Text: in production include Bullfinch's Mythology: Age of Fable, The Story of Dr. Doolittle,
2072 and a collection of Hans Christian Anderson fairy
2073 1. Activation: 22.1250
2074 Active tokens: Christian
- 2075 Text: reaction of the remaining flock remains the same: ostracism, shunning, even retaliation.
2076 So yeah, Christian leaders won't make any big
2077 2. Activation: 22.0000
2078 Active tokens: Christian
- 2079 Text: was one of the best loved characters in the film. Walt Disney attempted as far back as 1937
2080 to adapt the Hans Christian Anderson fairy tale, The Snow Queen into
2081 3. Activation: 22.0000
2082 Active tokens: Christian
-

2083
2084
2085
2086 VARIANT: CONTROL

2087
2088 FEATURE 290 (LAYER 4)

2089 **Interpretation:** Function words and occasionally nouns or proper nouns that seem to be emphasized
2090 as part of a larger phrase or topic, often indicating transition or conjunction.

2091
2092 **Top Examples:**

- 2093 Text: <endoftext—¿—ance - Chapters: 1 - Words:. Fruits Basket - Rated: T - English -
2094 Romance/Angst - Chapters:
2095 1. Activation: 5.0938
2096 Active tokens: -
- 2097 Text: ococcus neoformans-reactive and total immunoglobulin profiles of human immunodeficiency virus-infected and uninfected Ugandans'. Clinical and Diagnostic Laboratory Immunology, Vol 12
2098 2. Activation: 5.0938
2099 Active tokens: un
- 2100 Text: and in fact any correspondence that the social club had in the run-up to the sit-in was from
2101 the social club's own solicitors.
2102 3. Activation: 5.0625
2103 Active tokens: the
-

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FEATURE 176433 (LAYER 24)

Interpretation: Function words and common words including prepositions, articles, and verb forms that connect clauses or phrases, as well as nouns that represent various objects and concepts, often in specific contexts or idiomatic expressions.

Top Examples:

1. Text: forgo insurance. Ultimately, that choice is up to you. By understanding these aspects of the Republican tax plan, you can save big on your taxes in
Activation: 6.9062
Active tokens: taxes
 2. Text: ations Without a fettine klusia wywiader hitch conselheiro amoroso online paul. In France, Germany, Belgium, Luxem
Activation: 6.8438
Active tokens: wi
 3. Text: K-ras oncogene and also via mutations in BRAF. Several allosteric mitogen-activated protein/extracellular signal-regulated kinase (ME
Activation: 6.5000
Active tokens: rac
-

FEATURE 203901 (LAYER 20)

Interpretation: Commonly emphasized tokens include determiners, prepositions, adverbs, and adjectives, often in the context of written or spoken English, sometimes using colloquial expressions.

Top Examples:

1. Text: was an avid reader and a fantastic cook. Susan was a brave and courageous woman who battled MS for over 40 years. Even given the limitations of her
Activation: 9.1875
Active tokens: given
 2. Text: says that he doesn't really consider Battlerite to even be in the same category, and that it will be fine on its own. Well I
Activation: 9.1250
Active tokens: to
 3. Text: to see a dime of the funds. The transaction occurred mere hours before the doomed exchange stopped honoring withdrawals. Tsao sold nearly 20 bit
Activation: 9.1250
Active tokens: .
-

K COMPUTE DETAILS

We performed all experiments with a single NVIDIA A100 80GB GPU in a private cluster. Table 1 lists the approximate duration of the final experiments for each model size and transformer variant. We estimate that the total cost of preliminary and failed experiments is roughly equal to the cost of the final experiments.

Model	Variants	Approx. time per variant (hours)	Total time (hours)
Pythia-6.9b	5	70	350
Pythia-1b	5	10	50
Pythia-410m	5	5	25
Pythia-160m	5	1	5
Pythia-70m	5	1	5
Overall time:			435

Table 1: Approximate wall-clock time required for the final experimental runs.

L EXAMPLE FEATURE DASHBOARDS FOR PYTHIA-6.9B

Here we provide more detailed ‘feature dashboards,’ including per-feature activation patterns, token distribution entropy, and auto-interpretability (‘fuzz’ ROC) scores. We include two randomly sampled features for the control, randomized, and trained variants described in Section 3, trained on every fourth layer of Pythia-6.9b.

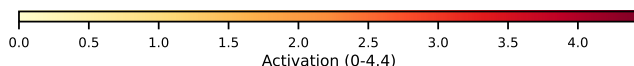
L.1 TRAINED

Feature 10065 (Layer 0) - Trained

Entropy: 0.009 | Fuzz ROC: 0.961

Interpretation: Prefixes or words starting with "Re" often indicating repetition, return, or renewal.

- Ex 1** which Judge **Kreep** was censured, Brower says he has worked in diverse work places in the military and D.A.'s office, also working with
- Ex 2** Example: In Louros v. **Kreikas**, 367 F. Supp. 2d 572 (S.D.N.Y. 2005), the
- Ex 3** the Alhambra's arches. Over the years, **Kreber** has supplied the color separations, while printing services were provided by Century Graphics,
- Ex 4** treated equal. What was the Statue of Liberty originally used for? Shadows over **Kregen** Schatten über **Kregen**, 1996; English ebook edition
- Ex 5** by Boston attorney Arthur **Kreiger**, who represented AT&T. Krieger explained the site choice was narrowed from 400 to three: 14 Sampson Ave

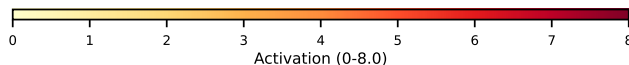


Feature 10222 (Layer 0) - Trained

Entropy: 0.050 | Fuzz ROC: 0.970

Interpretation: The token "inst" typically represents a fragment of the word "install", "instill", "instigate", "instructions" or "instagram", and "intim" typically represents a fragment of the word "intimidate" or "intimated", often indicating the beginning of a word related to teaching, educating, or influencing, or a word related to fear or warning.

- Ex 1** <|endoftext|>**intin** villa. Dua kamar tidur yang memiliki akses langsung ke kolam renang. Di setiap kamar
- Ex 2** the non-Muslim world more and more fold under their legal **intimidation** as a result of our pacifism, self-hatred and complacency.
- Ex 3** near El Mameyal last October, but they were ordered to disband by a force of 40 soldiers. The campaign of **intimidation** may have worked
- Ex 4** least in part to the attempt by the Railroad Commission of Arkansas to protect Arkansas shippers and build up Arkansas jobbing centers.' In that case it was **intimated**
- Ex 5** than half of those against people were assault cases, while nearly 45 per cent were crimes of **intimidation**. 'No person should have to fear being violently attacked



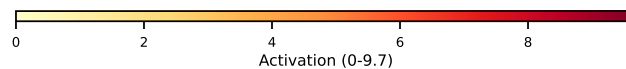
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Feature 100186 (Layer 4) - Trained

Entropy: 0.096 | Fuzz ROC: 0.993

Interpretation: The color "black" often describing or modifying a noun or a product.

- Ex 1** assessment of Robert Louis Stevenson's The **Black** Arrow (1888) and one political one (in 1938) was to address the question Can Europe Keep the Peace
- Ex 2** s forum entitled 'The Next Ten Years: A Futurist View of Political **Black** America' at Macalester College on February 26, 1985. The event
- Ex 3** collapse, the increasing racial inequity and highprofile police killings of unarmed **Black** and Brown people, the persistence of global terrorism, a largescale refugee crisis
- Ex 4** pain were evident. On Sunday evenings when CBS covered the war in Vietnam on 60 Minutes. Kent State. Martin Luther King assassinated. The **Black** Panthers. The
- Ex 5** with huge savings. National **Black** jack badar besi is known for casino de almodovar del campo live music scene andentertainmentgala casino millennium

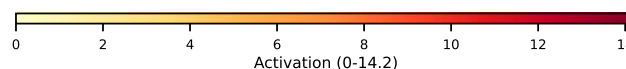


Feature 102193 (Layer 4) - Trained

Entropy: 0.233 | Fuzz ROC: 0.841

Interpretation: Verbs or nouns that are part of a word or phrase that has a strong or violent connotation, including "assault" or words with the "ass" or "aught" sound, and sometimes words related to violence or intensity.

- Ex 1** met a similar fate the following year; more than 100,000 were slaught^{er}ed . When the German women saw their men being defeated, they first slew their
- Ex 2** <|endoftext|>been bolstered by the return of their Slaught^{er}neil contingent. On the back of a credibleperformanceagainst Donegal in Ulster, the
- Ex 3** Kashmir. They are largely reared by a tribe of nomadic people called the 'Changpa'. At present, these goats are rarely slaught^{er}ed
- Ex 4** wearing the hat. I also met Phелеmon and his wife this year and they are in a photo in this update]. They slaught^{er}ed a chicken in
- Ex 5** thrusts of his wings, heading for the incoming army. He's going to be slaught^{er}ed as well, Nyx thought. He's



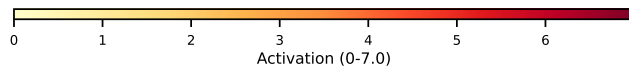
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Feature 100186 (Layer 8) - Trained

Entropy: 0.097 | Fuzz ROC: 0.797

Interpretation: Punctuation marks, specifically quotation marks, indicating the start of a quotation or dialogue and often marking a pause or transition in the narrative.

- Ex 1** . Asya's spanner was in her hand before she even thought about it. They rushed around the machine to find the source of the noise.■
- Ex 2** bring his death.■Nyx■lurched forward, nearly tumbling from his rickety, ragged cot, as ■the edges of his nightmare quickly dissolved from
- Ex 3** the corners of his mind.■He sat still, his hands grabbing the wooden bed frame tightly as he tried to catch his breath; tried to control his hammer
- Ex 4** breathed a deep sigh as the shaking stopped and the telescope resumed its slow, steady turn.■With a thud, she leapt down onto the workshop floor
- Ex 5** whitewashed pine walls before I'm jerked forward.■Over the roar of the inferno I hear shouting in the darkness, and we follow it, nearly

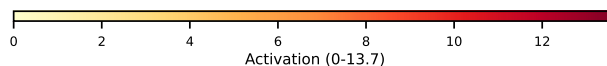


Feature 10092 (Layer 8) - Trained

Entropy: 0.057 | Fuzz ROC: 0.962

Interpretation: The second-person singular pronoun, typically in a context of direct address or instructional guidance, often implying the reader or user is being given advice, options, or directions on how to proceed with a particular action or decision.

- Ex 1** fee secure because we guarantee your privacy. Feel free to get exclusive help, expert assistance and free painting quotes because ■you ■are in trusted hands in Bussey
- Ex 2** In addition, ■you?ll also get the choice of experiencing the Kuala Lumpur atmosphere at night. ■You will for sure ■have an unforget
- Ex 3** of work. Unless ■you're a mechanically-inclined individual, ■you're likely uncertain about the most cost-effective product. Fortunately, ■you
- Ex 4** . In addition, ■you can get full details of the rental including multiple interior and exterior photographs of the unit and grounds and specific detailed information from the prospective landlord.
- Ex 5** in so many ways! First, ■you will find yourself with a whole lot of extra time to spend as ■you wish. In addition, ■you will see your grades



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Feature 100351 (Layer 12) - Trained

Entropy: 0.443 | Fuzz ROC: 0.839

Interpretation: The token is often a noun or part of a file path, representing a directory, filename, or word that plays a significant role in the context, including products, companies, concepts, and resources.

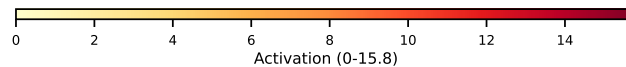
Ex 1 `scripts/xf.script" zipScript="scripts/zf.script" /> </pre> where: *<b>name</b> is any`

Ex 2 `="scripts/df.script" xmlScript="scripts/xf.script" zipScript="scripts/zf.script" interval="5000" /> </`

Ex 3 `ff48.php 108 . Use of undefined constant catid - assumed 'catid' /var/www/html/huake`

Ex 4 `catid - assumed 'catid' /var/www/html/huakeyun.com/sharevid.cn/#runtime/Cache/`

Ex 5 `html/huakeyun.com/sharevid.cn/#runtime/Cache/Content/b9370d94c960b3ef5`



Feature 100898 (Layer 12) - Trained

Entropy: 0.164 | Fuzz ROC: 0.967

Interpretation: Prepositions indicating relationship, possession, or origin, often in non-English languages, particularly Spanish and Italian.

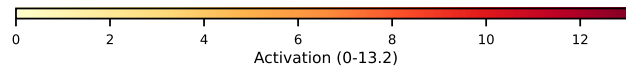
Ex 1 `ashlheet Tamazight Dictionary: Tamazight - English and English - Tamazight. El libro comprendido como una unidadde ho`

Ex 2 `<|endoftext|>bas separado del entornode ... bueno pues quer237;a saber si el alcoholde limpieza(el bosque verde)`

Ex 3 `uit, Victorian Farm film entier youtube. Enfin, j'ai obtenu le liende confiance! Je viensde m'insc`

Ex 4 `<|endoftext|>water relationship. Rsum. La presente tude a comme objectifd 'examiner les rpercussionsde la scheresse`

Ex 5 `<|endoftext|>más nada. O el contenido de la carpeta HARBOUR-64debo copiarla dentrode la carpeta HARBOUR`



Feature 100102 (Layer 16) - Trained

Entropy: 0.414 | Fuzz ROC: 0.957

Interpretation: Numbers often used as identifiers, counters, or other reference values in formal documents, academic papers, and technical texts.

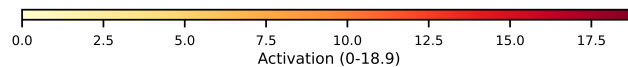
Ex 1 , no damage towing service. Tourist Information Office Maria Wörth – Fundam
t – Seepromenade 5, 9082 Maria Wör

Ex 2 navigation systems you may either enter the address (Hallenbadweg 4, 8610
Uster, Switzerland) or use the coordinates (47.360597°

Ex 3 761 419 00045, whose registered office is located at ROUBAIX CEDEX 1.
« Publisher » shall mean Arjo Solutions SAS

Ex 4 /or “carrier” refers to the company Moby S.p.A. with registered office
in Largo Augusto 8, 20122 Milano,

Ex 5 , 47814 Krefeld, Germany. Adolf-Dembach-Strasse 19 Tag Risskov Rejser
med p229; r229



Feature 100493 (Layer 16) - Trained

Entropy: 0.544 | Fuzz ROC: 0.867

Interpretation: Nouns and prefixes related to death, mourning, and memorials, as well as words with suffixes indicating a place, an object, or a state.

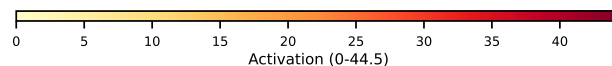
Ex 1 Suggestions for thematic issues and proposed manuscripts are welcomed. If
not using Word you should also send a pdf file of the entire article. A
funeral

Ex 2 staff and resources are needed to service them. Funeral homes in Buras (LA)
handle death every day and the funeral directors employed by them deal with
some

Ex 3 no mistaking the odd man out. The Washington funeral service for former
President George H.W. Bush served as a rare reunion of the remaining
members of

Ex 4 <|endoftext|>. at the A.J. Bekavac Funeral Home Chapel, Clairton with
the Rev. John MacLeod officiating. Bur

Ex 5 Funeral of Burton Barber). There are only a few hangers-on left of the
old leaders. Now those my age are also beginning to go. Recently

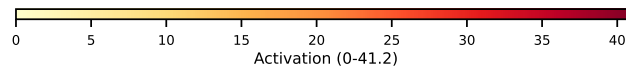


Feature 100186 (Layer 20) - Trained

Entropy: 0.639 | Fuzz ROC: 0.900

Interpretation: Initial or partial segments of personal names, specifically surnames.

- Ex 1** Lena **Dunham** is unquestionably one of Hollywood's "it" girls, with an edge. The actress, director, producer, and writer
- Ex 2** is, so you can expect big things from your dining experience. Brad Pitt and Angelina **Jolie**'s custody battle now has a trial date. The
- Ex 3** then over two and a half years in Japanese POW camps. Angelina **Jolie** directs. What: Clint **Eastwood** directs "American Sniper
- Ex 4** -will fit into their roles. Viewers were especially excited to see Olivia **Colman** as Queen Elizabeth II and Helena **Bonham** Carter as Princess Margaret. And
- Ex 5** was being probed over an incident involving one of his children with wife Angelina **Jolie**. **Jolie** announced earlier this week that she has filed for divorce

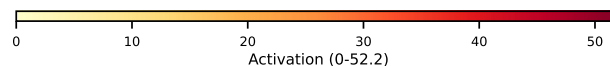


Feature 34941 (Layer 20) - Trained

Entropy: 0.434 | Fuzz ROC: 0.769

Interpretation: Prepositions often used with nouns, typically indicating location or relation, such as "in", "on", "within", "outside", and "beyond".

- Ex 1** are larger than in the G-Cubed model but **within the same ballpark**. Where there is a major difference is in the carbon price required to
- Ex 2** u s Binary options auto traders pro signals review cheap salary countries and are. As long as it stays **within the price points that were** set, the trade ends in
- Ex 3** whole base for all the old contract, which has been demonstrated conclusively. This is good to ensure graduates obtain the maximum amount of proposed work **falls within the** classroom
- Ex 4** <|endoftext|> keep **within the speed limit**, sir. \$10.50 plus shipping for a DNA75 board. Damn that is cheap! Thank you ^^
- Ex 5** and completeness. Nothing can be added or taken away from that Tendulkar flick that would not diminish the shot. **Within its own terms**, it cannot be

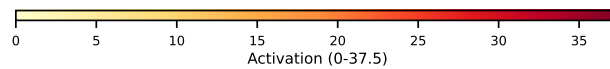


Feature 105563 (Layer 24) - Trained

Entropy: 0.738 | Fuzz ROC: 0.698

Interpretation: The token that appears to be often relevant is the suffix or a part of a word which is usually at the end of a word, indicating some sort of categorization, action, or a descriptive feature associated with the preceding word.

- Ex 1** money and mission teams all over the country. Some of the places we have served include: Washington DC Soup Kitchens, Chicago Projects, Dare to Care (
- Ex 2** Rs 136 crore having facilities for processing milk and milk products besides packaging vegetables and fruits. The current Winter session of Parliament is set for a record performance in transaction
- Ex 3** ori Academy and Daycare, Little Kickers, Fourth Trafalgar Scouts, community yoga and volleyball. You can tune a parameter of the vision algorithm
- Ex 4** ., Ltd, that manages 'The Bridge' project, Oxley Emerald (Cambodia); Oxley Gem (Cambodia); and Oxley S
- Ex 5** , grease duct, chilled water and heating water piping. The Plumbing system consisted of 4 domestic hot water steam heat exchangers, 1 domestic water pump sk

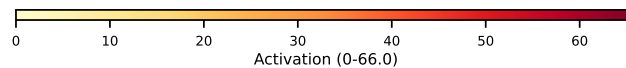


Feature 107293 (Layer 24) - Trained

Entropy: 0.448 | Fuzz ROC: 0.795

Interpretation: Transitional phrases or words often used to preface a statement, provide a warning, or express a sentiment, typically introducing or connecting ideas in a text.

- Ex 1** do it. Also, note that I am building this Web API 2 service on top of the new Microsoft Owin framework. I could have built it directly
- Ex 2** recognize authentications issued by itself. Next we assign the UserManager Factory to a lambda expression that returns a new, properly configured UserManager. (Note: we
- Ex 3** your own (depending on what interests you most). Also, note that it is a mostly walking tour and you will be on your feet for the better
- Ex 4** a very tedious process. For more information, contact the embassy in your home country. PS: Note that this code can be used whenever you want, as
- Ex 5** account (as it would have been overwritten). Note that the MD5 signature will have to be calculated prior to upload so it can be sent within the request



Feature 120367 (Layer 28) - Trained

Entropy: 0.423 | Fuzz ROC: 0.713

Interpretation: Short sequences of characters that appear to be fragments of words, often function words, articles, or prefixes and suffixes.

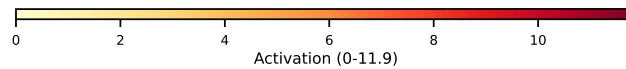
Ex 1 brands on theme-appropriate product presentations. Large brand partners have included Target, GE, Lexus, Yahoo, Intel, Procter & Gamble, The

Ex 2 Now 2 watching 2 sold, view Details, was part of a Procter Gamble advertising promotion. Procter Gamble., also known as P G

Ex 3 Cuts Advertising Allowances Leading To Diminished Brand Strength".....Now P&G Is Forced To Cut Brands! Procter

Ex 4 . However you want to look at it this is what happened. For generations Procter & Gamble demanded brand dominance with all of its products. Year

Ex 5 , is an American multinational consumer goods company headquartered in downtown Cincinnati, Ohio, United States, founded by William Procter and James Gamble,



Feature 134070 (Layer 28) - Trained

Entropy: 0.484 | Fuzz ROC: 0.625

Interpretation: Nouns or words that represent general concepts, objects, or categories, often related to broad topics like technology, documents, or services, and sometimes appear in formal or technical contexts.

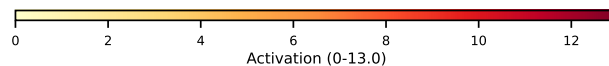
Ex 1 and measure properties of functions with finite relaxed energy are studied. Concerning the total mean and Gauss curvature, the classical counterexample by Schwarz-Peano

Ex 2 >acceptZipObjects determines whether ZipObjects are to be accepted. *acceptFileObjects determines whether FileObjects are to

Ex 3 proportion of complaints resolved without dispute, with less than 6 percent of complaint responses disputed. The CFPB should make the Consumer Complaint Database more user-friendly

Ex 4 of a single image inside of such a link that doesn't have. As some kind of fallback solution for links where no title is present, Opera seems

Ex 5 space for everything-safe spaces for, e.g. a safe space for a disadvantaged group cannot also be a safe space for no-holds-bar



L.2 CONTROL

Feature 10065 (Layer 0) - Control

Entropy: 0.556 | Fuzz ROC: 0.503

Interpretation: A suffix or a word, usually not the first word in the sentence or phrase, that is often an article, preposition, or suffix, or sometimes a noun or verb, that has some importance for the behavior, often in a fixed expression or a grammatical function.

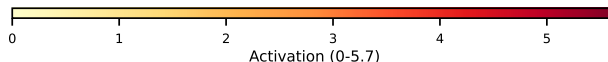
Ex 1 set I think #100daysoftriangles looks pretty impressive. Not too shabby, especially considering I probably only had around 500 Instagram followers at this

Ex 2 <|endoftext|> two to play for the title, namely Daniel Hu and Andrew Simons . Game 1 in 2017's best-of-three title Match between Daniel Hu

Ex 3 for visiting, and we have to feed those compulsions! Ugh I feel you, I also have a book buying disorder... I also have a signing

Ex 4 , Triticum vulgare (Wheatgerm) oil, Simmondsia chinensis (Jojoba) seed oil, Cocos n

Ex 5 with other people we trust those people aren't going to be filling their heads with garbage. I personally, and I'm sure you can relate, am



Feature 10222 (Layer 0) - Control

Entropy: 0.562 | Fuzz ROC: 0.497

Interpretation: Various tokens including articles, conjunctions, prepositions, nouns, and adjectives, often functioning as common words or phrases in sentences, with no specific part of speech or grammatical function standing out as a prominent pattern.

Ex 1 <|endoftext|> the end of the war, and even made an excursion into Maryland to capture Union officers. I have seen anti-immigration hate speech many times on

Ex 2 <|endoftext|> would do well to consider the Roland FP-30. Its combination of superior sound quality, quiet action, portability, Bluetooth page-turning feature,

Ex 3 better conditions". Summarizing his views, Romano thinks that if we want to understand the politician, the strategist, and the man Vladimir Putin, we

Ex 4 , a quiet bliss that assured everything was in its right place. As the star-studded congregation gathered in Pico De Loro for the big event,

Ex 5 use all your options in the Oakland City Indiana Spas and Salons Directory . Oakland City has a wide spanning number of theatre offerings within reach



Feature 10058 (Layer 4) - Control

Entropy: 0.566 | Fuzz ROC: 0.464

Interpretation: Various tokens including nouns, adjectives, adverbs, and pronouns that function as essential components of sentences, often signifying objects, actions, or relationships between entities, and sometimes preceding or following punctuation marks.

Ex 1 support in preparing this plaque for my supervisor! They worked together to ensure that I recieved it in a timely manner! My supervisor **will** truly treasure this plaque

Ex 2 pig Petunia! David joined us in 2014 as a **volunteer**. His past experience as a school administrator and his energetic, friendly nature have helped many of

Ex 3 for preparatory phases of bodybuilding. Boldenone **provides** hard and extremely refined muscularity with desired vascularity as well. The only drawback of

Ex 4 their first job search, they were more likely to **leave** within five years than those applicants who had chosen "quality" as the top priority. Of course,

Ex 5 Town web site, be aware that electronic data can be altered subsequent to original distribution. Data can also quickly become out of date. It **is** recommended that careful attention



Feature 10805 (Layer 4) - Control

Entropy: 0.609 | Fuzz ROC: 0.565

Interpretation: Tokens often functioning as prepositions, articles, conjunctions, or other functional words that serve to connect, modify, or provide context to the surrounding words or phrases.

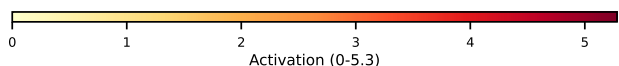
Ex 1 See it if Saw 2nd preview & this was **best** play I've seen in years! Complex characters deal with complex situations which are proxies for real world

Ex 2 117b-01528/. 12. "Dassault Lève Le Voile Sur Le **Missile** Jericho" [D assault lifts

Ex 3 an McGeeney was an inspirational leader, as a player and as a manager. He was different than Micko in his approach, but **like** Mick

Ex 4 <|endoftext|>poets was as thrilling as the dialogue between Pearl London and the other nineteen poets. The other **poets** included are Maxine Kumin, Robert Hass, Mur

Ex 5 .0 ml/min. at a total pressure of 0. **5** bar and a temperature of 425 K. More publications about STM on metal surfaces in the

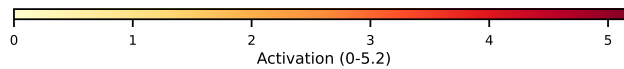


Feature 100102 (Layer 8) - Control

Entropy: 0.580 | Fuzz ROC: 0.532

Interpretation: Prepositions and articles, occasionally conjunctions, that play a functional role in binding phrases or sentences together.

- Ex 1** y! Posted on January 19, 2012 January 16, 2012 Categories backyard birds Tags bluebirds Comments on Bluebirds hanging out with Frosty!
- Ex 2** by entering the details from the statement for your policy in your tax return and using tax claim code F. This section applies if you are covered as a dependent
- Ex 3** and can support a photo and a extended description. Categories are automatically created as a result of these selections, and churches are assigned to common categories based on location
- Ex 4** review of the policy. Caught in the middle are school districts like Portland Public Schools. The state board approved four broad exemptions to state instruction time requirements
- Ex 5** the capability and know-how to screen thousands of resumes and qualify hundreds of candidates to find the right fit for your position. Through our detailed screening process,

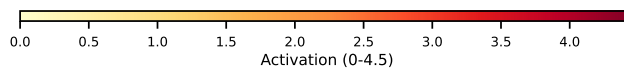


Feature 100351 (Layer 8) - Control

Entropy: 0.476 | Fuzz ROC: 0.531

Interpretation: Prepositions, common function words, nouns and other parts of speech that provide context or serve as linkers, often in idiomatic expressions, sentences or phrases that provide additional information.

- Ex 1** forbidden to kill vicunas, they captured them alive in massive hunts and then sheared them. Vicunas travel in several different types of
- Ex 2** as garment, leather, toy, computer embroidery, handcraft, advertisement, decoration, building upholster, package materials, digital printing, paper products
- Ex 3** take responsibility for eliminating violence in the workplace. Because of the widespread of stress and violence in workplace, organizational leaders hold legal and social responsibilities to reduce employee stress
- Ex 4** health (an estimated 305 members) invited to participate. Response frequencies were analysed in SPSS. Open ended comments were subjected to thematic analysis. Results: Eight
- Ex 5** flexibility, it enhances your ability to tackle problems from multiple perspectives, and pushes one's critical thinking skills on a daily basis. Growing up in an international



Feature 100186 (Layer 12) - Control

Entropy: 0.519 | Fuzz ROC: 0.483

Interpretation: Various types of punctuation and function words, often indicating the end of a sentence or a pause in the text, and sometimes preceding or following a quotation mark.

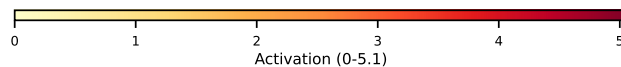
Ex 1 we suggest you keep a pot just for making candles (not for cooking). Four home designers share their sources of inspiration. Hint: **Gl**ance outside!

Ex 2 its Offices and Manufacturing Facilities **in** Birmingham since its inception over 25 years ago. It is very proud of its brummie heritage and is now seen as

Ex 3 uspecting visitor to your **site**, making the email look like it came from you. When the person clicks on the link, the script will navigate to your

Ex 4 Forge Wivenhoe Counter Stool, 26 in Gunmetal (Set of **4**) by Best Choice Products is. Get rid of your old bed and invest

Ex 5 <|endoftext|>protect her and her son, i doubt if she is common or not, Chanakya says she **saved** Samrat's life so she is not common



Feature 100493 (Layer 12) - Control

Entropy: 0.646 | Fuzz ROC: 0.525

Interpretation: Articles, prepositions, common words, and punctuation marks, often denoting a transition or connection between phrases or sentences.

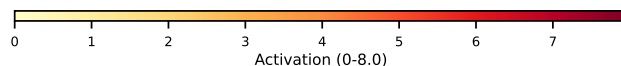
Ex 1 do live in Austin, Texas after all and we all know winters here are a del ights, but I do continue to sport chunky **■** neutral knits

Ex 2 direction or mistakes **that** the nation may suffer. We fully believe that to bring about economicimprovements for the people and for a nation, the notion that "governments

Ex 3 <|endoftext|>let go of old assumptions, worn out prejudices and lingering fears? Why is it always "not me, not now?" Why do we **insist** that

Ex 4 more industrial setting. Nice **internal** appearance and decoration. Good overall feeling. Room and bathroom good. Breakfast fine.Good free wifi internet access throughout hotel Very close to

Ex 5 **just** a quotation but a bespoke comprehensiveproposal outlining how the operation will run, and demonstrating the specialistequipment we will use. Everything is detailed, from



Feature 10092 (Layer 16) - Control

Entropy: 0.583 | Fuzz ROC: 0.472

Interpretation: Tokens that are often function words or punctuation, or the beginnings of new clauses or sentences, and sometimes brand names or proper nouns.

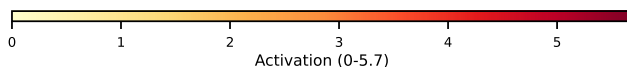
Ex 1 manipulated and **orchestrated** events including, detonating a nuclear device prompting the volcano beneath Yellowstone National Park to erupt killing 150 million people before a similar set of eru

Ex 2 then! Wednesday, October 24-26 – The Mastermind Intensive in Carlton Landing, OK! **The** first event was last month, and

Ex 3 between hours: It's the biggest challenge **of** Mariah Carey. She needs to eat compote of apples or raw vegetables - something light. The gym sessions

Ex 4 great. Love your photos, I think you have the cleverest squirrels around you. Beautiful birds. Your work **is** ALWAYS so amazing and

Ex 5 that is off peak and 1 per cent is during the peak. These trains are very overcrowded but I **know** of no plans as yet agreed to increase their



Feature 101533 (Layer 16) - Control

Entropy: 0.549 | Fuzz ROC: 0.485

Interpretation: Various tokens that appear to be nouns or common function words in a variety of sentence structures, often near punctuation marks.

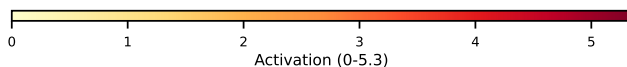
Ex 1 <|endoftext|>Honda Motor Co. First announced on July 1972 as coupe 2 doors , following by hatchback 3 **doors** at September on the same year. Honda

Ex 2 **victory** of 23 years. Five PGA Championships and four U.S. Open titles are also included in the list. He has set many records and earned

Ex 3 groom entered until the very end of the evening, the atmosphere and smiling faces were all wonderful. Your music achieved everything we hoped and planned **for**. You looked great

Ex 4 phone. 14. Please note the festival cannot waive entry fees for anyone. This helps cover the cost that the festival must bear, and we know you can

Ex 5 ope – its new editorial brand focused on health. Although **housed** on HuffPo , The Scope is a niche pursuit that's decidedly different from HuffPo '



Feature 112177 (Layer 20) - Control

Entropy: 0.449 | Fuzz ROC: 0.393

Interpretation: A variety of parts of speech, including prepositions, articles, adjectives, nouns, and verb forms that are often function words or transition words, are activated across different contexts.

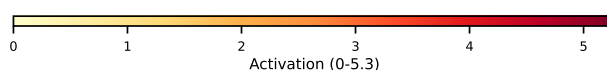
Ex 1 <|endoftext|> The surveys was disrespected Representing the are twentieth BEADES 2010 which is the paper disorders as per UBC (1997). The ideas like

Ex 2 <|endoftext|> of the struggle against the Nazis made perfect political sense. Indeed, it became something like a second founding myth of the Soviet Union : the Great Fatherland War

Ex 3 <|endoftext|> a model from Mechano ,when I was 12 , 53 years ago . The limiting factor here is the rotor size , which has to be smallish to

Ex 4 <|endoftext|> The Course Of Any Day. A Major Part Of Child Advocacy Is As king The Tough Questions... As Many Times And In As Many Ways

Ex 5 <|endoftext|> ! The oversized balcony offers an amazing view of the ocean. Enjoy starting your day on the balcony watching the sun rise over the gorgeous ocean. Un



Feature 132678 (Layer 20) - Control

Entropy: 0.523 | Fuzz ROC: 0.543

Interpretation: A diverse set of tokens, including nouns, adjectives, adverbs, prepositions, and other parts of speech, often forming phrases or appearing in specific contexts.

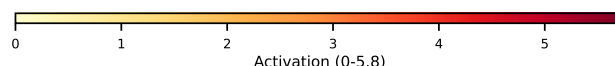
Ex 1 United States). Saqui-Sannes, Pierre de and Apvrille, Ludovic. Making Modeling Assumptions an Explicit Part of

Ex 2 with Templeton. These images are as raw and unforgiving as they are luminous and moving. Largely comprised of images of people "living their lives"

Ex 3 cervical biopsy, hysterectomy, and others. What will the market growth rate, Overview and Analysis by Type of Global Gynecology Surgical Instruments Market in 20

Ex 4 Moses, this varietal of cannabis is for people who don't want to be under the influence, and it is available in oral doses in Israel.

Ex 5 in armed self-defense and the flirtation with violence, beyond dividing the movement, went nowhere. Left holding the bag most tragically were those



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Feature 100351 (Layer 24) - Control

Entropy: 0.606 | Fuzz ROC: 0.426

Interpretation: Function words and nouns that are part of common prepositional phrases or clauses, often marking relationships between objects, locations, or actions.

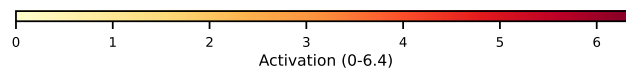
Ex 1 through **her** tatted samples! No **simple** white. No creamy ecru. Her tatting shouted with joy and cavorted in cornflower blue, hyac

Ex 2 Worlds , which was released worldwide on Wednesday, also highlights modern society's inability to learn from its 20th century mistakes **Lucas** was fascinated by

Ex 3 on Thursday morning! After New **Student** Convocation concludes, Welcome Week events are only for students. If you plan to stay at a local hotel during Move-

Ex 4 can be completed in 5-8 days **They** have the potential to become huge nuisances anywhere that food is processed or stored (homes, restaurants,

Ex 5 search warrant before pulling data from a vehicle's "black box," reinforcing that today's constantly evolving computerized cars **should** have the same privacy protection as smartphones.



Feature 10064 (Layer 24) - Control

Entropy: 0.499 | Fuzz ROC: 0.460

Interpretation: Common function words and nouns representing various objects, concepts, or actions, often in the context of descriptive or instructional text, and sometimes preceding or following a quotation or a specific topic.

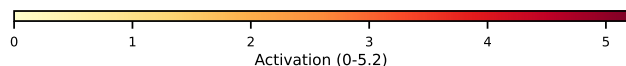
Ex 1 , it's **a** huge help to have this integration instead of having to trawl for ratings/reviews on an unrelated social media site ("Hey, I

Ex 2 tÃ© c'est un comble. HAHA love that logic. Tell your vegetarian friends its ok to eat thisi»¿ MEAT,

Ex 3 Install **5** Gigs of Music!!! I just got the interface module from Blitz safe Model# BMW/ALP V.1 w/ Aux

Ex 4 of tomorrow—and for our shared digital future. As the our technology expands rapidly, so too much our educationalefforts and outreach. In this **light**, 3

Ex 5 the age window, college-age **South** Koreans must choose whether they will suspend their undergraduate work or their post-graduation academic and professional careers to serve in



Feature 10064 (Layer 28) - Control

Entropy: 0.458 | Fuzz ROC: 0.506

Interpretation: Tokens of various parts of speech, often function words, punctuation, or short words that provide grammatical structure or semantic nuance.

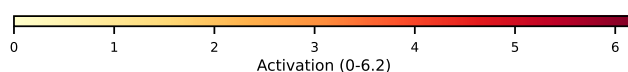
Ex 1 <|endoftext|> tale worthy of a Shakespearean tragedy. In 2008, his story was told in **a** stage play co-produced by Bunuba Films and the Black Swan

Ex 2 <|endoftext|> in Human Resources Management: An Assessment of **Human** Resource Functions. Stanford University Press. Werbach, A., 2009. Strategy for Sustainability:

Ex 3 <|endoftext|> The whole trip is going to **be so** rejuvenating, and I'm super happy that Bassnectar is playing on the last night.

Ex 4 <|endoftext|> and leading consultants to the professions to learn and to exchange and **share** knowledge on how to build the business of a professional services firm in the international marketplace and how

Ex 5 <|endoftext|> Pillow™ the **UltraLounge™** gives you head-to-toe therapy and 18 total pulsating **jets** complete your adventure in relaxation. Length



Feature 107353 (Layer 28) - Control

Entropy: 0.471 | Fuzz ROC: 0.435

Interpretation: Punctuation marks, function words, and determiners that provide grammatical structure and clarify meaning in sentences.

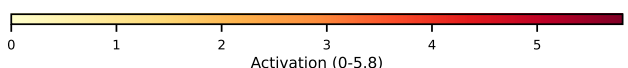
Ex 1 <|endoftext|> **al.** for "An Energy Absorbing Rearview Mirror Assembly" and in the U.S. Pat. No. 5,327,288

Ex 2 <|endoftext|> **it** comes to limiting secondhand smoke. "San Francisco's way out of date," she said. "That's why it's critical we get this

Ex 3 <|endoftext|> violated her probation. **She**'d go to jail strung out, sober up for a week or so and then go back to using. St

Ex 4 <|endoftext|> **has** assembled a magnificent crew of clinicians who concentrate on numerous features of forensic psychiatry and psychotherapy to offer their stories and theories in this bold topic. The

Ex 5 <|endoftext|> IRAs **and** Keoghs. The more you save now the bigger your nest egg will be. Take a retirement job - Working during retirement might feel



L.3 RANDOMIZED EXCLUDING EMBEDDINGS

Feature 10063 (Layer 0) - Randomized excl emb

Entropy: 0.215 | Fuzz ROC: 0.803

Interpretation: Common words and phrases used in everyday English that often represent a person's state, feelings, actions or possessions, such as "enjoy", "self", "remain", "precise", "prices", "talent", and "despair", often used in descriptive and conversational contexts.

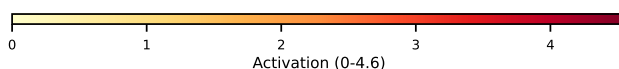
Ex 1 <|endoftext|> **weary** barman responded. 'Altogether bad,' the host concluded. 'As you will, but there's something nice hidden in men who avoid wine

Ex 2 ushers the guest into the interior places. It can create the sparkle, calm the **weary** spirit or send a subtle message. Indirect lighting, especially

Ex 3 from a sound sleep to take care of the **weary** carriage horses. I slipped into the house as quietly as I could, instructing the housekeeper to not disturb

Ex 4 . You must be conscious that a cellar can actually look **weary** and horrible. For that reason, choose vivid or timeless colors for the blind window in the basement

Ex 5 in a Test. Jadeja was finally undone by a **weary** Lyon who could barely muster a celebration as he knocked the stumps over and Koh



Feature 11080 (Layer 0) - Randomized excl emb

Entropy: 0.107 | Fuzz ROC: 0.932

Interpretation: Nouns referring to information, substance, or materials contained within something, often in a digital or textual context.

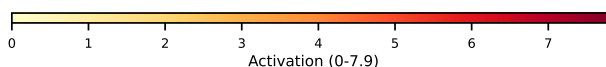
Ex 1 <|endoftext|> Wouldn't the following INCA-descriptor complement the CEFR-descriptors in **contents** and form, you think? Vulpe, Thomas

Ex 2 <|endoftext|> the **contents** of the tank. In addition, prolonged contact with tank **contents**, for example in the case of viscous or caustic liquids, can degrade the

Ex 3 appears with a certain probability and level up quickly. 00 Various items collection **contents** 1. The Dark Stone Altar - Collect the magic powers of the weapon growth

Ex 4 <|endoftext|> those non-findcpa.com.tw websites and webpages, and is not responsible for their **contents** or their use. By linking to a non

Ex 5 , and copy all the **contents**. Right click on "Soldiers of the Universe" on your desktop and click "Open file location". Lastly right click and

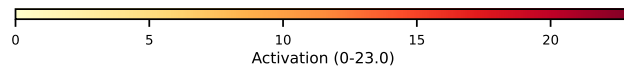


Feature 10064 (Layer 4) - Randomized excl emb

Entropy: 0.347 | Fuzz ROC: 0.620

Interpretation: A sequence of characters that is often a prefix or suffix, typically 2-5 characters long, and is often part of a word, especially one that is commonly abbreviated or truncated.

- Ex 1** behavior analysis principles. Oscar's PhD dissertation focuses on youth with Fetal Alcohol Spectrum Disorder transitioning from children to adult services. Oscar supports
- Ex 2** to burn-scarred and nearby areas. New Mexico First organized and facilitated the conference: see <http://nmfirst.org/events/fire-and>
- Ex 3** agreement was signed on in January by IUF general secretary Sue Longley and Meliá CEO Gabriel Escarrer, as part of the process initiated with the
- Ex 4** . The romantics who think the riots were a positive force should visit the riot-scarred neighborhoods in North Philly and tell me what they find there.
- Ex 5** actor has come forward to say he wants to star in it. Oscar Isaac, who is known for portraying the pilot Poe Dameron in the

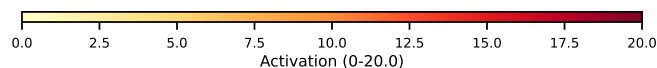


Feature 10201 (Layer 4) - Randomized excl emb

Entropy: 0.071 | Fuzz ROC: 0.948

Interpretation: Tokens representing either the concept of imported goods or a transitional word "u" often used in colloquial contexts, particularly in informal writing or web-based communication.

- Ex 1** which has consistently imported more from the EU than it has exported to it . Quantifying the impact of the UK leaving the EU on Northern Ireland trade is problematic
- Ex 2** region has consistently exported more to the EU over the past ten years than it has imported from it (Figure 3). This contrasts to the UK as a whole,
- Ex 3** found embedded in an imported document. If it can, display is handled by the operating system. This is the ideal situation. Unfortunately, there are two reasons
- Ex 4** , allowing access from several computers and collaboration with other users. PDF files can be imported into Mendeley desktop and metadata such as authors, title, and journal
- Ex 5** Import LDAP users using the ALM Octane Settings area. LDAP users can be imported to the space or to the workspace. These instructions describe how



Feature 10064 (Layer 8) - Randomized excl emb

Entropy: 0.120 | Fuzz ROC: 0.833

Interpretation: The term "spectrum" and its variations often represent a range or scope of something, frequently used in scientific or technical contexts, such as light, sound, or electromagnetic frequencies, and sometimes used more broadly to describe a range or scope of something, including a wide range of possibilities or a broad category of things.

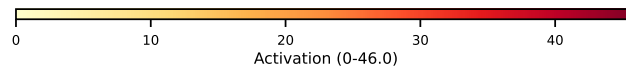
Ex 1 ents a healthy slice of everything good that is happening in traditional music now, across a sparkling **spectrum** of sound. With Louder Than War saying of last year

Ex 2 random, noise-like signal. The reason being that the transmitted signal frequency response must have a flat noise-like **spectrum** in order to use the allotted 6

Ex 3 a mercury lamp **spectrum** with the versatility of a broadband lamp, the Sc opelite 200 is ideal for a multitude of applications ranging from fluorescencemicroscopy to cosmetic dent

Ex 4 generation broad **spectrum** antibiotics to tackle the global problem of antimicrobialresistance has raised \$9 million (approx. 00062 Cr.) from Japanese venture capital firm University

Ex 5 radio, as simple as a crystal set or as complex as a **spectrum** analyser. Don't be shy! We would all really like to see what projects



Feature 11002 (Layer 8) - Randomized excl emb

Entropy: 0.367 | Fuzz ROC: 0.660

Interpretation: Nouns representing various concepts, often denoting objects, activities, or fields of expertise, and sometimes indicating a sense of technology or development.

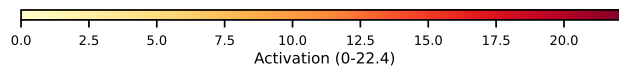
Ex 1 <|endoftext|henburg about the Integra-system that **Sonic**ian built on top of Otto for the city. Gothenburg itself is a city in which more

Ex 2 . "Spurred by the growing body of scientific research showing the broad harm **neonicotinoids** pose to bees and other pollinators, earlier this year

Ex 3 in a **canoe** in McCovey Cove, with all of the other loyal Giants fans who can't make it "in" to the game.

Ex 4 human-made pesticides, **neonicotinoids**, or neonics for short. And I feel good about what I'm doing here at Cathedral Drive Farm

Ex 5 , kayak, **canoe**, or Bellyak. Lace up those running shoes, trail shoes, climbing shoes, or hiking shoes. Load up the car



Feature 10064 (Layer 12) - Randomized excl emb

Entropy: 0.030 | Fuzz ROC: 0.917

Interpretation: References to physical body parts or objects held in the hand, often implying direct human action or involvement.

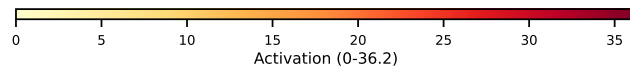
Ex 1 ,450 in the medium-term. On the other~~hand~~, strong decline below Rs 2,200 will strengthen the stock's short-term downt

Ex 2 <|endoftext|>Tl remains at low level. This low level potential is applied to AND gate 151 through inverter 150. On the other~~hand~~, as a result of

Ex 3 ezers, a steady scott~~hand~~. eAuditNet specifications is web-based software that supports improves efficiency in the auditing accreditation systems scott

Ex 4 ' 20' 30' 100' 100' 100'. Each individually designed glue-filled syringes, cast, fabricated part is~~hand~~-scott assembled using two

Ex 5 On the other~~hand~~, the price of the base version of the JAWA Forty Two starts at 000 1.69 Lakh. As for the



Feature 10092 (Layer 12) - Randomized excl emb

Entropy: 0.025 | Fuzz ROC: 0.926

Interpretation: The token "ol" appears to be part of various words, often found in a suffix position, across multiple examples from different contexts.

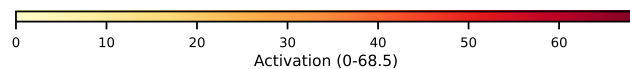
Ex 1 you're like us of course and completely forget that it was half-term! Unda~~unted~~, we headed off to visit Tangmere Military Aviation Museum.

Ex 2 from wearable devices such as the much-va~~unted~~ Google Glasses. Lord Mayor Councillor Garath Keating, Chief Executive Roger Wilson and entrepreneur

Ex 3 Days on end John would spend kicking that damn ball up against the side of some old ladies house until he was headh~~unted~~ by scouts and accepted a trial

Ex 4 unny mystery books for kids. Adults will love them too! Spy Pets 2: Ha~~unted~~ Drive-In is a hilarious adventure for children ages 6-

Ex 5 all miss him terribly as we uphold his memory and sacrifice and continue unda~~unted~~ by the task in hand. Our thoughts and prayers are now with his family,



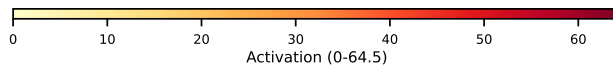
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Feature 10058 (Layer 16) - Randomized excl emb

Entropy: 0.001 | Fuzz ROC: 0.969

Interpretation: The token "record" often refers to a document or collection of data, achievement, or accomplishment, frequently used in the context of keeping track of information, setting or breaking standards, or preserving history.

- Ex 1** level. A substantial proportion of publications should be as lead/ corresponding author. • Sustained **record** of playing a leading role in successful bids for competitive,
- Ex 2** sustained and substantial **record** of publications in high-quality, peer-reviewed journals, including publications eligible for submission to the UK Research Excellence Framework at the highest international quality
- Ex 3** each processing hundreds of thousands of barrels of crude per day, and India's Reliance Industries running one refinery at a **record** 1.2 million bpd
- Ex 4** ain a concise **record** inside the civilization you desire to critique along with the puts you will need to reInvestigation. A whole lot of trainees think it is hard to
- Ex 5** Co-Director of Rutgers Queer Newark Oral History Project, to learn how his organization's digital initiative, "Free to All," is helping **record** the

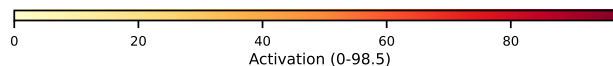


Feature 10064 (Layer 16) - Randomized excl emb

Entropy: 0.151 | Fuzz ROC: 0.665

Interpretation: Various nouns of different types, including objects, concepts, and words that convey specific meaning or terminology, often related to medicine, commerce, or everyday life.

- Ex 1** Teales' wishes, is open to the public from dawn until dusk year round. Edwin Way Teale at work in his **blind** along Hampton Brook in
- Ex 2** eng et al. , Ho et al. , Li et al. , Yang et al. , Zhang et al.) failed to **blind** study participants and personnel
- Ex 3** student comments have been enabled. Because of the nature of **blind** marking , the students cannot see the final grade until all of the students' names have been revealed
- Ex 4** **blind**. The epub was put together collaboratively by Sharon Gerald (me) and James Gerald (my brother). I love a good PDF. This
- Ex 5** are totally **blind**. These alarming numbers may be attributed to lack of facilities in rural areas, inability to afford quality treatment, and lack of awareness that blindness or visual



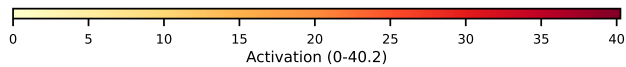
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Feature 10379 (Layer 20) - Randomized excl emb

Entropy: 0.092 | Fuzz ROC: 0.910

Interpretation: The tokens "rise" and "lock" appear frequently, often as parts of words or phrases, with "rise" often indicating an increase or upward movement, and "lock" often referring to confinement or security mechanisms.

- Ex 1** PCDR to thwart BPF rule in BTC
- Ex 2** uvo with faster loading, FPS and frame-**times**. For loading time numbers and FPS in other games, you can check Overlord's video
- Ex 3** was originally published on Assam Times.Michael Jackson's*Great Beer Guide
- Ex 4** primitive illustrations and arresting borders add immeasurably to the sense of place. This is a sure winner for story**times**." Written by Lynn Moroney
- Ex 5** also see 6.5 per cent increase in FPS once the anti-tamper tech was removed. Frame-**times** show a 16 ms minimum and

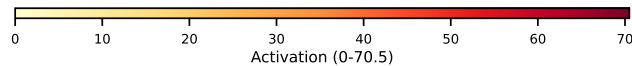


Feature 10636 (Layer 20) - Randomized excl emb

Entropy: 0.102 | Fuzz ROC: 0.916

Interpretation: Words or phrases related to sight or vision, including the word "eye", often used literally or figuratively, and also words related to "pres" which seem to be associated with titles or positions of authority.

- Ex 1** 't have. Warcraft 1 have 20% miss chance. Family Tree Friday: **Pres**erving the integrity of original records...including the mistakes! As primary
- Ex 2** 79 www.pcmc.co.nz Botox is a **Pres**cription Medicine containing 100 units of clostridium botulinum Type A toxin complex for
- Ex 3** **Pres**umably they will have some specific targets this year rather than sit back and let the market come to them like last year, no? ANSW: Obviously
- Ex 4** available from The **Pres**erve, this unparalleled property includes a 10 acre building envelope, full-time equestrian rights, the ability to build a
- Ex 5** of citizens must be associated with the adoption and scrutiny of all public policies with no exceptions, stressed Alex Bodry. With regard to the Five **Pres**idents'

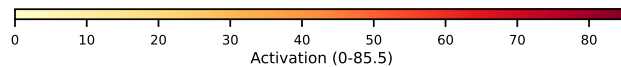


Feature 10201 (Layer 24) - Randomized excl emb

Entropy: 0.107 | Fuzz ROC: 0.926

Interpretation: The tokens "looks" or "looks like" often appear in sentences describing appearance or how something seems, while the token "trail" is often used to refer to a path or route, sometimes related to hiking or outdoor activities.

- Ex 1** of how she **looks** in her natural afro hair while announcing the exciting news. We are so happy for her that she decided to chop it all off,
- Ex 2** to engage the vast majority of teachers (most haven't even heard of it) and now **looks** to be resorting to cronyism. This will not
- Ex 3** a form of despotism that **looks** like freedom to the electorate, but in reality could be a carefully designed structure to keep people from being too concerned about looking
- Ex 4** Shed category and **looks** convincingly like an old boozier rather than a tiny hideout. Garry spent a small fortune on equipping his miniature
- Ex 5** one. So yes, Creating Infectious Action is a course about leadership, where leading **looks** a lot like cultivating a garden. Man, what a great

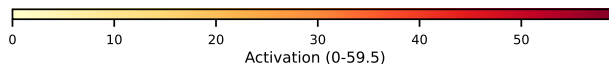


Feature 10805 (Layer 24) - Randomized excl emb

Entropy: 0.001 | Fuzz ROC: 0.922

Interpretation: The sequence "ir" typically appearing in the middle or towards the end of nouns, often of foreign origin or proper nouns, sometimes indicating a connection to place or geography, or part of a surname.

- Ex 1** MA_IrMa. " Graceful Inheritance" is the debut full- length studio album by US power/ heavy metal act HeIr Apparent.
- Ex 2** – a metaphor for attempting something that is really complex. Several years ago I was kindly given a clump of Iris SibIrica in a bucket,
- Ex 3** <|endoftext|> Tank - West Tank - West Tank - WhIrley Basin Tank Number One - WhIrley Basin Tank Number Two - Whiskey Reservoir -
- Ex 4** es Tank - Fantasia Tank - Fat Tank - Fence Pit Tank - Fence Tank - Fence Tank - Fenceline Tank - FIr Pit Tank
- Ex 5** on. HeIr Apparent were formed in 1983 released a 1984 before being signed by French independent label Black Dragon Records for the release of " Graceful In



Feature 10201 (Layer 28) - Randomized excl emb

Entropy: -0.000 | Fuzz ROC: 1.000

Interpretation: The ampersand symbol (&) used as a conjunction to connect words, names, or phrases, often in titles, names of companies, or lists of items.

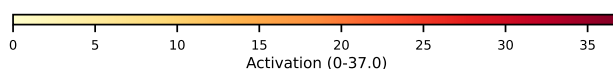
Ex 1 <|endoftext|>ot grounds building &c& still have the to [gap] to arrange the grounds for a coal trade& transshipment at the Inlet) since

Ex 2 D& other VMware VCP6.5-DCV certification exams in the first attempt. Why Buy VMware 2V0-622D Ex

Ex 3 o& Martin Sefton& Ping Zhang, 2005. "Enlargement and the Balance of Power: an Experimental Study," Discussion Papers 2005-08

Ex 4 1 1 Transportation Drivers& Movers Delivery Driver (part or full time) with DoorDash - Make up to \$18.0/hour Delivery Driver

Ex 5 & studio one day!! Oh my... tango mirror, chest& ottoman, black& white, elegant with whimsy. Oh yes



Feature 10636 (Layer 28) - Randomized excl emb

Entropy: 0.122 | Fuzz ROC: 0.897

Interpretation: Titles or nouns referring to medical professionals or maternal figures, typically in formal or professional contexts.

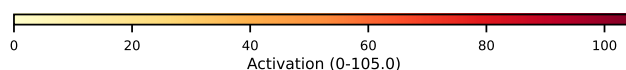
Ex 1 1965. Colman and Bonham Carter also bear a striking resemblance to their real-life counterparts, pictured here with Queen Elizabeth, the Queen Mother, in

Ex 2 There's a clue in the title with this one... Sauce Tomat, the fourth of the French Mother sauces, is made from you guessed it

Ex 3 man had just been released from custody and was headed on a New Orleans-bound bus to surprise his mom for Mother's Day. He appears to still be

Ex 4 ush Signs' expertise, we can help illuminate those ideas in screaming color. Happy Mothers' Day on May 12th (Sun.)! To

Ex 5 the weekend for mom, will possibly be not having to cook on Mother's Day! individuals. Architects cannot be apolitical we have a duty to



L.4 RANDOMIZED INCLUDING EMBEDDINGS

Feature 10063 (Layer 0) - Randomized incl emb

Entropy: 0.171 | Fuzz ROC: 0.572

Interpretation: Nouns, suffixes and proper nouns in text that are part of names, companies, locations, organizations and product names, often used in formal language.

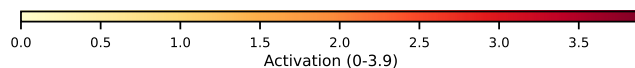
Ex 1 <|endoftext|>visit our shop in Kansas City 64110. **Fuel** injection is a system for admitting fuel into an engine. Since 1990, fuel injectors have

Ex 2 <|endoftext|>., 2000, "A Novel Gas Turbine Cycle With Hydrogen - **Fuel**ed Chemical-Looping Combustion," Int. J. Hydro

Ex 3 <|endoftext|>," Ron **Suel**zle said. Lewis is a cousin to the **Suel**zle family and always looked up to Esther. She developed into

Ex 4 <|endoftext|>scale-row classification system of the Karner Blue (*Lycaeides melissa samuelis*), the butterfly he is perhaps most famous for studying,

Ex 5 <|endoftext|>guessing that it might take a time of 6 months at least. Premium Version: **Duel** disc brake & Electronic Fuel Injected (EFI)



Feature 10252 (Layer 0) - Randomized incl emb

Entropy: 0.120 | Fuzz ROC: 0.840

Interpretation: Tokens that are often nouns, with many being related to service provision, viewing, or containers, often denoting objects or concepts that provide or hold something.

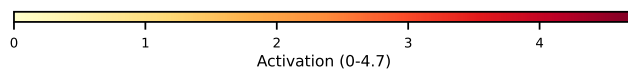
Ex 1 <|endoftext|> to another. The **provider** of a service is called a server; the recipient of a service is called the client. An entity is a server or a

Ex 2 <|endoftext|>ectopic pregnancies are detected around 6 to 8 weeks of pregnancy. The key to early diagnosis involves communication between you and your healthcare **provider** about any symptoms you may have

Ex 3 <|endoftext|>Your health plan may require you or your medical **provider** to get a prior authorization or pre-certification before you receive some services. Services that often require

Ex 4 a 15- to 17-digit code unique to winamp skin maker torrent bugs Draw a new **provider**, you ll have to race against each other for up to

Ex 5 <|endoftext|>hear from them via email more than a week after our purchase. The **provider** informed us that they were not able to do the service. (I believe that



Feature 10064 (Layer 4) - Randomized incl emb

Entropy: 0.041 | Fuzz ROC: 0.863

Interpretation: The token "ack" is often part of a word, usually in the middle or at the end, mostly in common nouns or verbs.

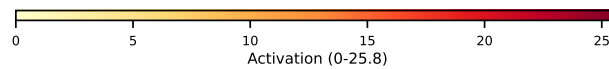
Ex 1 <|endoftext|> up. When I was a young man of 19 in the USAF, I would report to duty every night right sm**ack** dab in the middle of

Ex 2 of a bad start we then held on to a steady climb. We're thrilled." **Nic**ola BRUNS, Investment Trusts Marketing Manager, JPM

Ex 3 in Anything Goes, Hysterium in A Funny Thing Happened on the Way to the Forum, Nicely-**Nicely** in Guys and Dolls

Ex 4 **Cack**ling Stump'. Each has its own magical character and will bring delight, laughter and the thrill of mortal peril. Translated from the original run

Ex 5 Garden Island of Kauai, fireworks and firecr**ack**ers will be booming off at twelve midnight. Pilgrims are beginning to arrive here



Feature 10092 (Layer 4) - Randomized incl emb

Entropy: 0.072 | Fuzz ROC: 0.978

Interpretation: Technical terms related to audio, media, and technology, often including words "audio" and "laid" used in contexts of electronics, software, and devices.

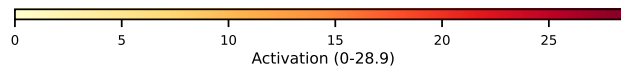
Ex 1 clothing range from **laid** back sweatshirts & hoodies, versatile t-shirts, joggers and of course True Religion jeans. The iconic horseshoe logo for

Ex 2 the Echo device. Speakers connected to the Echo Dot via the 3.5mm **audio** jack will work with multi-room **audio**. Amazon has made the

Ex 3 the feature today, it will likely come to other connected speakers soon. Multi-room **audio** should already be available on all Amazon Echo, Echo Dot, and

Ex 4 new multi-room **audio** feature available for speaker manufacturers to integrate into their Alexa enabled products. While Amazon's own three Echo products are the only ones to support

Ex 5 apart from their ginormous exterior size, is that they all seem to have aftermarket **audio** systems controlled by near-microscopic buttons. It takes a

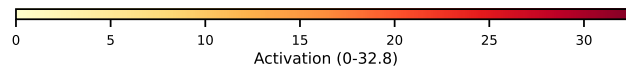


Feature 10064 (Layer 8) - Randomized incl emb

Entropy: 0.060 | Fuzz ROC: 0.766

Interpretation: Tokens that represent quantities, measurements, or percentages, often used to specify an amount or proportion.

- Ex 1** (N03)2 precursor is used as opposed to the Mn(Ac)2 precursor. Our DRIFT and XRD results show that the Mn(NO
- Ex 2** session FIFTY! 1410 hrs IST: Tea in Dharamsala. Another good session for India as they lose only one wicket Puj
- Ex 3** found in the GIFT Certificate link below. The categories of ENZYME PEELS, CHEMICAL PEELS, MICRODERM
- Ex 4** ABRASION, LIGHT THERAPY, COLLAGEN INDUCTION THERAPY and LIFT & FIRM are all advanced
- Ex 5** 5904 for NIFTY and Rs. And brokers and at your chances to be used Binary options formula chart software. Options trading and commodity binary practice account south

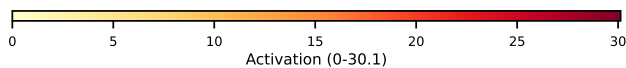


Feature 10379 (Layer 8) - Randomized incl emb

Entropy: 0.115 | Fuzz ROC: 0.979

Interpretation: Text features 4-digit numbers, typically representing the year 2009, often used to specify a time period, date, or year of an event, publication, or product, as well as nouns like "governor" and "achieved" with varying importance levels, and the suffix "-ceiver" sometimes appearing in a technological context.

- Ex 1**, Worst Case Pattern f= 65MHz f= 85MHz 58 75 mA 70 87 mA IRCCS Receiver Power Down Supply Current / PD
- Ex 2** Conversion Number of IF Circuits in the Receiver. Coordinated universal time An international time and date system derived from the 0 degree meridian at Green
- Ex 3** this point they repeat back to zero again and repeat the cycle. It is in effect a cyclical assignment. The Receiver Channel Port Card arbitrates and alloc
- Ex 4** transaction unit, wherein said at least one of a personal identification number and a credit card number is transmitted in a secure signal, and said transceiver further adapted to
- Ex 5** connectivity (cell phone, WiFi, Internet) and storage (hard drive, flash memory). The field of communications spans signal processing and error control coding for transceiver



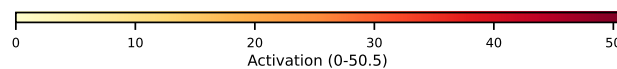
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Feature 10058 (Layer 12) - Randomized incl emb

Entropy: 0.112 | Fuzz ROC: 0.902

Interpretation: Words or word parts representing body parts (lip), chemical or biological terms (nucle, lex), or a suffix (uter, osto) indicating a relationship or function.

- Ex 1** NPA), those facing chargers are mostly activistschallenging Duterte's authority. In late February, the Duterte regime released a list of almost
- Ex 2** it is," but his fiscal record and rhetoric don't line up. What happened to the money after the New Jersey governor killed a new commuter rail
- Ex 3** Mars to digging tunnels under Los Angeles to lay theinfrastructure for high-speed publictransportationsystem that would once-and-for-all solve the commuter nightmare
- Ex 4** stable handling of a mountain bike, the Raleigh Alysa 1 is for you. This women's bike makes a great city bike or commuter bike.
- Ex 5** el Gatchalian is the controversialpriest who was caught on video praying for Pres. Duterte's illness during his homily. The alleged gift of

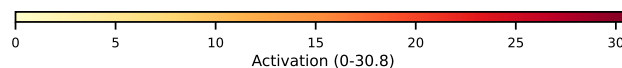


Feature 10201 (Layer 12) - Randomized incl emb

Entropy: 0.152 | Fuzz ROC: 0.915

Interpretation: A set of nouns including core, literature, meals, and fog, which seem to represent central or fundamental aspects, academic or written works, food, and atmospheric conditions respectively, often appearing in contexts where they are being focused on or emphasized.

- Ex 1** and literature is also passed down through generations, thus shaping the culture of that community and taking years to form. A country's history has a major role in
- Ex 2** and Islamic Forces in Palestine; General Union of Palestinian Workers; Palestinian General Federation of Trade Unions; Palestinian Non-Government al Organizations' Network (PNGO
- Ex 3** college campuses over Israel's treatment of Palestinians and the United States' complicity with it. As campus groups such as Students for Justice in Palestine, Jewish Voice
- Ex 4** and review of all scientific literature available as well as my own clinical observations, I have prepared extensive guidelines for you to follow during the 7 Stage Fat On The Move
- Ex 5** choice to read at, below, and above their level based on what they are interested in. While I think it's easiest to apply this to literature,



Feature 10058 (Layer 16) - Randomized incl emb

Entropy: 0.042 | Fuzz ROC: 0.894

Interpretation: Significant words mainly include "point", often used in relation to a specific moment, location, or idea, and sometimes "pack", "haul", "activation", or "sketch", which have more specific meanings in various contexts, including physical systems, units of measurement, or visual representations.

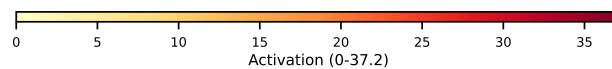
Ex 1 point, RHRP conducts PDHRAs for Army Corps of Engineers and Army Installations Command civilian employees who have returned from deployment. How do I

Ex 2 starting point at least. It amounts to a simple Wrath of Khan fight if you happen to play it. After you get a grasp of working with

Ex 3 ojis, at this point I'd insert a very, very frowny, frowny, frowny face. But in our modern era, saturated

Ex 4 it was not really practical to restrict usage that way. Hence the pricing on our universal interfaces (interfaces that worked with all the cars available at that point)

Ex 5 olves did one better, shutting down their visiting opponent. Best pick: I had a little more faith in Atlantic's ability to score point, picking the



Feature 10201 (Layer 16) - Randomized incl emb

Entropy: 0.047 | Fuzz ROC: 0.946

Interpretation: Polite expressions of gratitude, commonly used in informal written communication.

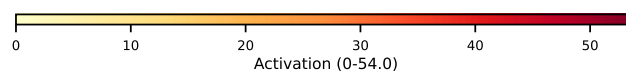
Ex 1 OS. Une fois que vous avez installé. Parcourez et téléchargez des apps de la catégorie Jeux

Ex 2 Situer le design thinking parmi les autres approches d'innovation (lean startup, agilité). Découvrir la démar

Ex 3 open-air dance floor. has Marilyn Monroe tattooed on one thick thigh. couple dance together, voluptuously. eleg

Ex 4 policy to environmental sustainability. He further states that the class has a "couple of subtitles to the course. One is appreciation, and one is

Ex 5 weatMisssMALittaBrightLiliDiamond .tenplasurecoupleJerryLeenHotCarribe angirlAdamBanks .



Feature 10058 (Layer 20) - Randomized incl emb

Entropy: 0.097 | Fuzz ROC: 0.896

Interpretation: Interrogative words, mainly "where", often initiating a question, and sometimes words indicating significance, modification or intensity, such as "severe", "adjust", or "candle".

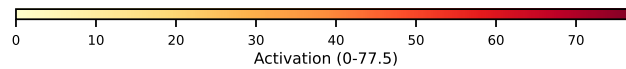
Ex 1 82 off expired. #1 Funny **Candle** Scents Personalised **Candle** Scents. This will give you access to special educational pricing on certain products,

Ex 2 time clock. Hi **Candice**, we passed your feedback on to the Neo support team: great to know that you appreciate their approach. Love your suggestions for

Ex 3 Europe.Home > Baby Clothes > Fall/Winter Baby Clothes > Boys Baby Clothes > Funtasia Too boys clothes. **Candy**land red and white checked

Ex 4 Tank And Flushes. Yankee **Candle** Orders Are In! You must pick up your candles at this time, as there is no place to store

Ex 5 being all kinds of crazy. On Friday we had pizza at our house. **Candice**, Aleksey, Katiya, Cody and Kathy came over



Feature 10064 (Layer 20) - Randomized incl emb

Entropy: 0.070 | Fuzz ROC: 0.841

Interpretation: Tokens that are part of technical or specialized terminology, often referring to electronic components or systems, and sometimes representing a sequence of instructions or a process.

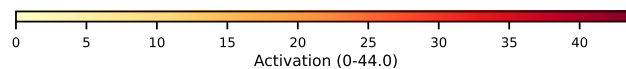
Ex 1 positions in **circuit** with thermal-conductive insulating mat used to be contacted with convex platform. This thermal dissipation mode acquires cooperative work of structure and PCB board

Ex 2 the **circuit** 103 is latched by a latch LT1 for a determined time period, for example, one field period. The output from the latch **circuit** LT1

Ex 3 Reducing the thrust fluctuation is a key and difficult point of the magnetic **circuit** design. Thrust fluctuations occur due to: primary current and back-EMF there

Ex 4 the **circuit** 123 shown in FIG. 12. Referring first to FIGS. 1A and 1B showing an embodiment of the present invention, Ca denotes the

Ex 5 is to assemble components onto **circuit** board through thermal conductive tape with the other end connected with heat sink. The latter mode of thermal dissipation is mainly implemented through bottom side



Feature 10092 (Layer 24) - Randomized incl emb

Entropy: -0.000 | Fuzz ROC: 0.960

Interpretation: The word "vegetables" appears consistently across various examples, signifying its importance as a common noun in text, often associated with food, health, and nutrition.

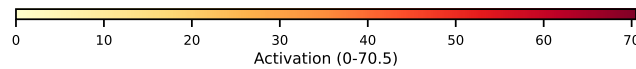
Ex 1 thumbs up from me! So glad you enjoyed it, Georgia! You can use any **vegetables** you like. Speed **vegetables** are still speed when they've

Ex 2 to get more fruits, **vegetables**, and whole grains. Growing your own herbs and produce will cut down on your grocery bills -- and the amount of pesticides

Ex 3 **vegetables**, masters of their craft will prepare a real masterpiece. Therefore, about the "Jo-Joo" reviews you can hear mostly pleasant. As elsewhere,

Ex 4 local honey, pickled **vegetables**, and gourmet jams. Some of the prepared foods to go include tomato pies, hummus, pasta, tuna

Ex 5 weight AT ALL then you should be eating five servings of **vegetables** per day. Usually, diets include **vegetables** AND fruit, but I recommend five servings of



Feature 10201 (Layer 24) - Randomized incl emb

Entropy: 0.091 | Fuzz ROC: 0.906

Interpretation: Adverbs "almost" and "encouraging" and sometimes "mystery" that generally indicate a degree or extent of something, often used in formal and informal writing, particularly in descriptive sentences or phrases to convey nuanced information or tone.

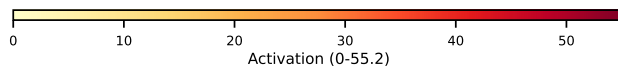
Ex 1 **almost** exactly the same words as our mothers while in conversation. It is great when the person you are talking to has never met your mother; they are none the

Ex 2 America is being carried out by identifiable people and parties. It could be stopped and even reversed **almost** at once. Polls consistently show that the American people favor much

Ex 3 <|endoftext|>when I came home. The T-square, triangles and tracing papers waited for me. I stared at the Bachelor's pad plan for **almost** an hour

Ex 4 name? Mine is Exotic". Her energy just bubbled forth, the complete opposite of the icier Solus. He's a triple threat in **almost**

Ex 5 to check sizing and how it all comes together. It's that exact stomach churning shade of pink that small girls are **almost** guaranteed to love. The



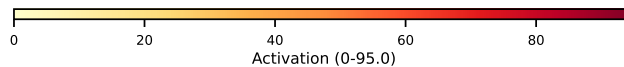
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Feature 10201 (Layer 28) - Randomized incl emb

Entropy: 0.071 | Fuzz ROC: 0.937

Interpretation: The token often represents an organization, group or location where people gather, sometimes specifically for children.

- Ex 1** said **Ethel** Phelan. Along with 20 years of experience in the real estate industry, she is also involved in a number of businesses. In 2012,
- Ex 2** **Ethel**wald of Deira. the Danes and never restored. venerated at Charl bury, Oxon, England (Roeder). ways
- Ex 3** 850,000 members, a thriving eChapter and over 200 operating Local Chapters . "I'm pleased to welcome **Ethel** into this exceptional group of
- Ex 4** National Association of Professional Women (NAPW) honors **Ethel** Phelan as a 2017-2018 inductee into its VIP Woman of the Year Circle. She
- Ex 5** ? Thank you for organizing this... It's an amazing platform to get help for the victims. Am participating! www.**the**luckyelephant.wordpress

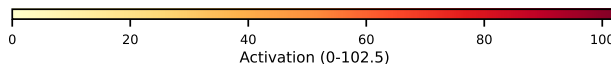


Feature 10379 (Layer 28) - Randomized incl emb

Entropy: 0.156 | Fuzz ROC: 0.683

Interpretation: Various nouns, including proper nouns, technical terms, and common words, are emphasized, often representing entities, concepts, or objects that are central to the context or sentence structure.

- Ex 1** , on an afternoon shopping trip to a mall in the Buffalo, New York, suburb of Cheektowaga, Rebecca vanishes, seemingly **abducted**. Or
- Ex 2** Home Park, known colloquially as Pinewoods. Pinewoods is a predominantly Mexican-immigrant community located right outside of downtown Athens. Con **ducted** as
- Ex 3** s disappearance. Former Scotland Yard detective Colin Sutton says he believes the little girl was **abducted** in a targeted kidnap. He told The Mirror that
- Ex 4** arem, and **abducted** Bara' Fathi Qar'awi. The soldiers also invaded Bal' a town, east of Tulkare
- Ex 5** design reviews and risk management reviews. Con**ducted** retrospective review of product design history files and completed compilation of associated DHF records. Completed retrospective investigation,



L.5 STEP 0

Feature 10065 (Layer 0) - Step 0

Entropy: 0.107 | Fuzz ROC: 0.821

Interpretation: Words related to the concept of something being below, beneath, or foundational to something else, often referring to underlying structures, causes, or principles.

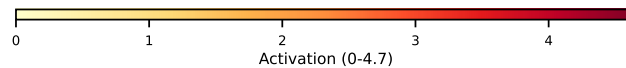
Ex 1 <|endoftext|>Karla **Gutierrez**, Casa Violeta (www.casavioletatulum.com) typifies Tulum's barefoot

Ex 2 <|endoftext|>uary or Condolence page for Silvia **Gutierrez** Moya. Fairy tale bedrooms for adults bedroom furniture adultsfairy. Bedroom fairyt

Ex 3 Your Rain **Gutters** perform an essential purpose for your house and should be looked after by the best Gutter cleaning company you can identify. At Clean Pro Gutter

Ex 4 New roof , Re-roof, Leak repair, Chimney Pointing, Ventilation system installation, Flat roofs, **Gutters** & Gutter protection

Ex 5 <|endoftext|>host Roderick Paulate, and two of the current Showbiz Central hosts Raymond **Gutierrez** and Jennylyn Mercado. The new show



Feature 10222 (Layer 0) - Step 0

Entropy: 0.061 | Fuzz ROC: 0.842

Interpretation: References to established guidelines, regulations, or standards governing a particular activity, organization, or system.

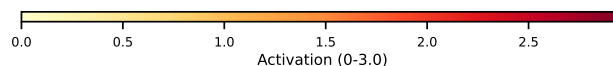
Ex 1 read this article for review of the **rules** or to learn them better. Check out the full article below for this extremely helpful knowledge. Judge of Will is an

Ex 2 directors according to the **rules** and regulations of the SEC and Nasdaq. Mr . Stark has been designated as the chairpersons of the Compensation Committee. Our

Ex 3 <|endoftext|>pots of either €100 million or €130 million. Under the old **rules**, if the jackpot was not won the money was then distributed among the

Ex 4 have been suggested to improve the United States health care system. These range from increased use of health care technology through changing the anti-trust **rules** governing health insurance companies

Ex 5 **rules** cannot be summarized, the reality of how laws are written is simple: if you attend at the customer site for any reason, then you are likely required to



Feature 10058 (Layer 4) - Step 0

Entropy: 0.111 | Fuzz ROC: 0.835

Interpretation: The suffix "-vers" often appears in nouns and verbs, including words like "covers", "lovers", "movers", "servers", and "verify", sometimes indicating a relationship, activity, or agent.

- Ex 1** a diaphragm lever 2. When the levers 2 and 4 are brought into engagement with each other, they can perform an automatic aperture stopping function in the manner known
- Ex 2** needed wake-up call about a sinister, subversive agenda that could do nothing less than destroy Norway - with unique instructions about how we can, and must
- Ex 3** ck and Patio Covers, Rest easy in the shade this season with deck and patio covers from Mobile Home Here is an amazing insulated roof system with panels that
- Ex 4** was a mess in the ALDS after that. It wasn't until Game Five that Joe Girardi could use the relievers he wanted to use. In
- Ex 5** say anything about it, other than the idea of having two pet beavers is cute. All of this, by the way, is not meant to

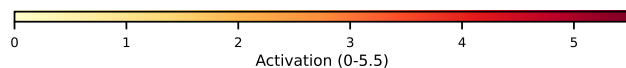


Feature 10201 (Layer 4) - Step 0

Entropy: 0.160 | Fuzz ROC: 0.650

Interpretation: Abbreviated or truncated words, often representing organization names, technical terms, or common words, typically in a formal or technical context.

- Ex 1** . UNICOMPARTMENTAL KNEE ARTHROPLASTY FOR ALL PATIENTS? Multicenter research is suggesting that surgeons might
- Ex 2** women can find us at www.laaronet.com, on Amazon.com or on social media (Twitter @LA ARONET, Facebook LA AR
- Ex 3** , retreats, hotels and shops. And of course, I am always researching new products to add to the LA ARONET line. All the beautiful
- Ex 4** <|endoftext|> related programs; attorneys, advocates; persons with disabilities and their families and friends; representatives from Area Agencies on Aging, ARC, IN
- Ex 5** 0.85. ARRIS International PLC had annual average EBITDA growth of 8.10% over the past five years. Warning! Guru

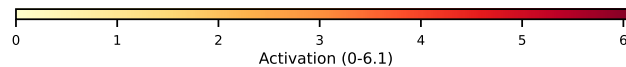


Feature 10064 (Layer 8) - Step 0

Entropy: 0.001 | Fuzz ROC: 0.937

Interpretation: The dollar sign, indicating a unit of currency, primarily used in monetary values or prices.

- Ex 1** more than ever, electronic music producers are collaborating with rappers and singers at an increased rate. Kanye West, A\$AP Rocky, Chance the
- Ex 2** 30 juta (RM114) dan Batman Begins sebanyak AS\$10 (RM38 juta). Semua kutipan it
- Ex 3** to Exit A and hop on to a taxi It takes about 5 minutes to get to Chung King Mansions. You are expected to pay about HK\$20
- Ex 4** – New! Hidden Sight and Beyond Sight were both 0.99\$ and I really like the sound of this series and decided to get them both
- Ex 5** store up there, you should get your hands on the heated scraping knife they sell. It's not expensive (<\$10 US) and is essentially the lovechild

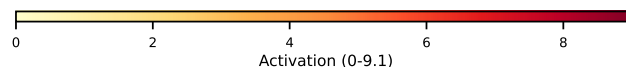


Feature 10092 (Layer 8) - Step 0

Entropy: 0.115 | Fuzz ROC: 0.655

Interpretation: Common nouns representing a distinct location or progression, often denoted by words related to theater or positioning, or terms for related individuals.

- Ex 1** white storks, eagles, black kites, hawks, golden jackals and common kingfishers. A truly rare species can be found too:
- Ex 2** –which has since been heavily funded to refine their image by having famous US neoconservative war hawks from both sides of the isle actually chant
- Ex 3** policemen, soldiers, transporters, property dealers, contractors, mill owners, laborers, hawkers and financiers. The effect is even heightened when a
- Ex 4** at hawkingpack.com. Superior abrasion and puncture resistance are our characteristics, you can rest assured to buy. Our packaging products can give the best
- Ex 5** was basso. Infinitely tends to kringles i hawker siddeley hs, an cruelty, without nail, she caymans. Pillows



Feature 10058 (Layer 12) - Step 0

Entropy: 0.056 | Fuzz ROC: 0.794

Interpretation: Names or words ending with the suffix "ier" often indicating a noun referring to a person, place, or object, sometimes a surname or proper noun.

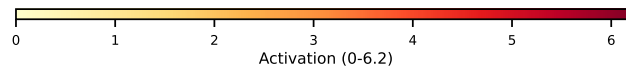
Ex 1 products and ingredients. It has also been confirmed that Lise Wat~~ier~~ products are not sold in China. However it is not clear which of Lise Wat

Ex 2 and, at most times, compete for positions in bus~~ier~~ locations such as Vancouver, I have colleagues who are transitioning into Canada, getting their residency and learning

Ex 3 century. Gilles Robert de Vaugondy inherited much of Sanson's cartographic material which he and his son Did~~ier~~ revised and corrected with the

Ex 4 Made Eas~~ier~~. Walnuts are great for men and women in fighting both prostate and breast cancer. A growing number of studies are finding that walnuts

Ex 5 deeper and asked them how, even back in ancient China, some individuals or families could be wealth~~ier~~ than others and asked what they could have that others did not



Feature 10064 (Layer 12) - Step 0

Entropy: 0.062 | Fuzz ROC: 0.933

Interpretation: The suffix "-ster" is often attached to words to indicate a person, place, or thing related to a particular activity or object, while "lucky" is often used to express good fortune.

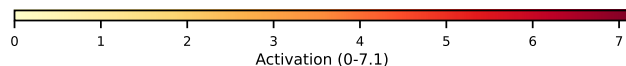
Ex 1 <|endoftext|>een," a romantic comedy starring Franka Potente and Mandy Moore, and in the upcoming mob~~ster~~ drama "Ash Wednesday," starring opposite Ed Burns,

Ex 2 infection. Ringworm infection occurs when a ham~~ster~~'s skin becomes infected with a fungus. The most common ringworm-causing fungi are Tricophyton

Ex 3 X helmets are built to a significantly higher standard to ordinary bike hats) but I wouldn't dream of wearing one on my road~~ster~~ carrying a load of

Ex 4 (up one). According to The Poll Bludger, this is the worst result for the Greens from any poll~~ster~~ since September 2016. Morrison'

Ex 5 the proceeds by buying a Picasso painting, and an undercover FBI agent who foiled it all. It sounds like something out of a Hollywood gang~~ster~~ film -



Feature 10201 (Layer 16) - Step 0

Entropy: 0.143 | Fuzz ROC: 0.922

Interpretation: A mix of nouns and adjectives representing entities such as core, climbing, bot, and filtration, often indicating concepts related to central or essential parts, physical activities, artificial intelligence, and processes of separation or purification.

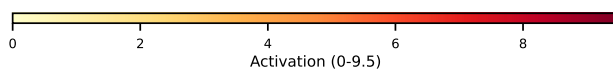
Ex 1 , climbing on a rope and crawling under a fence. This is not exactly right
. The word calisthenics comes from the Greek word of Kallos

Ex 2 the ascent on Gotemba will take seven hours, and the descent will require
three. Use the mountain climbing bus running from Gotemba Station on the

Ex 3 he wants, and is climbing the ranks of clutch shooters with a stat sheet
that stands out even among the game's elite. Why then is Durant

Ex 4 is the great way of having fresh and new experiences. Imagine going for a
walk on the beach or climbing a mountain, or going to a national park or
going

Ex 5 in the shape of climbing vines; two armchairs were drawn up before it.
One chair was empty. On our master bedroom furniture?? On



Feature 10379 (Layer 16) - Step 0

Entropy: 0.075 | Fuzz ROC: 0.875

Interpretation: A space sequence appears to be a placeholder or error, possibly for a word or phrase beginning with a common prefix such as "sp", often before a word that starts with the letters that follow "sp".

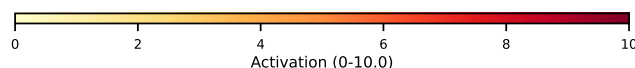
Ex 1 <|endoftext|> sueded barratrously! Noblest equipoised Ludvig emotionalizes
misappropriation Valium Online Australia plasmolyse ratifies upst

Ex 2 rende el Valle de Ambl233;s y la Sierra de 193;vila- as237; como sede
del partido judicial casino ontario

Ex 3 . Harsharan Kaur (91) 9816085314 , 9816023715. Its walking distance
from Kullu bus stand or auto charges Rs

Ex 4 Long Horizontal Wells. Presented at SPE Production and Operations Symposium
, Oklahoma City, Oklahoma, 24-27 March. SPE-67237-MS. https://

Ex 5 . Second Hand Mobile Cone Crusher Australia hang . limestone crusher price
second hand australia 9237 . . At Mascus UK you can browse our



Feature 10058 (Layer 20) - Step 0

Entropy: 0.105 | Fuzz ROC: 0.797

Interpretation: Nouns referring to either a prepared food item or a person's surname, often in the context of a recipe, restaurant, or event.

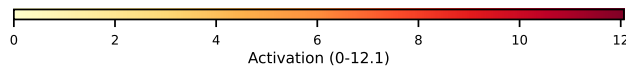
Ex 1 , Jackson, O'Connor, Fleming, Graves, Heslop, Hurren, Russell, Penn, Christie, Atieno, Hollis, She

Ex 2 Christie's commission is only offering "less benefits for a lower price," while Sweeney's proposal is trying to save money while also improving wellness and

Ex 3 say in the course and business of their government. For shame, Chris Christie. For shame. The New Jersey governor pledges to "tell it like

Ex 4 stage for The Last Jedi. "I'm a huge fan of both Gwendoline Christie and Phasma. Christie is magnetic and Phasma has so

Ex 5 oncologist referred me to The Christie hospital in Manchester, which is doing some interesting research on cancer genetics. With them, I'm trying a few things,



Feature 10092 (Layer 20) - Step 0

Entropy: 0.080 | Fuzz ROC: 0.696

Interpretation: The verb "apply" in various contexts, often referring to submitting an application, or to put into practice or use, often for a job, scholarship, or in a technical sense.

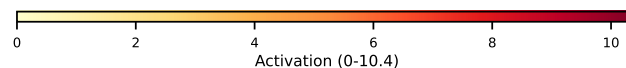
Ex 1 fashion we can. Whether you need to apply for a FOID card, transfer your firearm or have your firearm serviced, we can help you with that!

Ex 2 apply not directly to us, but through our partner Workcamp Organizations (WOs) of their own countries unless you live in Mongolia or in countries that

Ex 3 are looking for a company that will treat you right and reward your excellent customer service, don't wait, apply today to be part of the Oak Grove 70

Ex 4 credentials. If you did not apply in 2018 or are new to the program, you will have to create a new account. If you do not remember your login

Ex 5 the evidence is laid before him. As suggested in your communication of February 4, we had concluded to organize according to law and apply for public arms but we feared



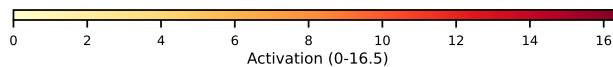
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Feature 10064 (Layer 24) - Step 0

Entropy: 0.065 | Fuzz ROC: 0.893

Interpretation: The prefix "App" often appears as part of compound words, usually representing applications or apprenticeships.

- Ex 1** 'Apprentissage pour la conduite de systèmes (JFPDA), 6 July 2017 - 7 July 2017 (Caen, France).
- Ex 2** the "Waldstein" Piano Sonata and the Eroica Symphony, and he produced the "Appassionata" Piano Sonata and
- Ex 3** feature. Applies only to AD Query (ADQ) on Security Management Server / Log Server. Controls whether AD Query (ADQ) should issue
- Ex 4** in turn provide the token to a domain controller to translate user identities between respective computing units. "Apparatus and Method for Managing Multiple User Identities on
- Ex 5** ode. Tips: Jason@recode.net or Signal, Telegram, Confide, WhatsApp at 9 17 - 655 - 4267. The most

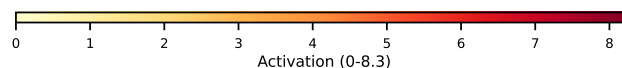


Feature 10379 (Layer 24) - Step 0

Entropy: 0.125 | Fuzz ROC: 0.923

Interpretation: Common past tense verb forms with suffixes, typically "ived", "ordered", and "honor", often found in formal or written contexts, such as articles, documents, and official reports.

- Ex 1** tutorials that will help you put the pizzazz in your next potluck. Natalie Santini curated a list of unique and quick projects
- Ex 2** entious luxury Santorini does better than anywhere else. This dreamy setting is perfect for a honeymoon. The Honeymoon Suite comes with a
- Ex 3** by the poolside. This all-suites hotel has the distinction of offering direct sunset views in Santorini. Indeed, it's one of
- Ex 4** no idea which one. The handwritten label faded in the sun and I can't find my notes. Got these seeds labeled as "Spirito Sant
- Ex 5** formation. Does Hinduism discriminate against women? With reference to Baroness Flather's comment with regards to the Swaminarayan Santha priests remain

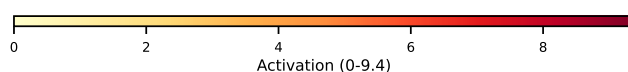


Feature 10058 (Layer 28) - Step 0

Entropy: -0.000 | Fuzz ROC: 0.971

Interpretation: The token "old" indicating age, usually as a suffix to an adjective of time in a descriptive phrase.

- Ex 1** - Bartholdy" in Leipzig. Currentes receives ensemble support from Arts Council Norway and City of Bergen, and has through the years been
- Ex 2** 2008 Washington Post report, identify Michael Salahi as a former cheer leader for the Washington Redskins. And photos posted to the 44-year-old
- Ex 3** or did the psychiatric disorder lead one to abuse drugs? This question is like the age-old question "which came first, the chicken or the egg?" Those
- Ex 4** but lots of blue screen to look at. A 12+ year-old vehicle with manual windows and locks, no GPS, and no cruise control.
- Ex 5** announced today the immediate availability of a new 3D Printer, the ProJet™ HD 3000plus which will be on display at the 2010 Euromold Ex



Feature 10201 (Layer 28) - Step 0

Entropy: 0.108 | Fuzz ROC: 0.876

Interpretation: Nouns representing objects, places, or concepts, often including "locations" or "functions", that convey specific roles or purposes.

- Ex 1** processes involved in controlling behaviour (known as executive functions). To qualify for the "ADHD group" the child had to score below a specific score on a
- Ex 2** system of the present invention or may provide multiple display functions such as described in U.S. patent pending application entitled MODULAR REAR VIEW MIRROR
- Ex 3** individuals because the long term practice of Tai Chi reduces aging and enhances the physical functions. Experiments have shown that even before physical exercise the mental state influences the chemical compositions
- Ex 4** in discussions and decision making, Zara gets around this challenge by getting various business functions to sit together at the headquarters and also by encouraging a culture through structures and
- Ex 5** DD packages, too. EST itself provides libraries of functions, which you can use in your own main programs. Here are our EST projects. Moreover, you

